

**UNIVERSITÀ DEGLI STUDI DI
MILANO-BICOCCA**

**CORSO DI LAUREA IN
SCIENZE STATISTICHE ED ECONOMICHE**

ANNO ACCADEMICO 2024/2025



State Space Methods for Electoral Polling

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Acknowledgements

Desidero innanzitutto esprimere la mia più profonda gratitudine ai miei genitori, che con il loro sostegno emotivo, fisico ed economico hanno reso possibile ogni passo di questo percorso. Nei primi anni di università, quando i dubbi e le incertezze erano più forti, siete stati il mio punto di riferimento: vi ho chiesto consiglio tante volte e la vostra presenza mi ha dato serenità e forza. Avete saputo accompagnarmi con fiducia e rispetto, lasciandomi sempre libero di scegliere la mia strada e dimostrando di credere in me. Siete il mio porto sicuro sia nei momenti di felicità sia nei periodi di preoccupazione, vi voglio bene. La mia riconoscenza va anche a tutti i miei familiari, che con affetto e incoraggiamento non mi hanno mai fatto mancare vicinanza e calore. In particolare ai miei nonni a cui sono tanto legato e che ringrazio per tutto l'affetto e l'amore che sento quando li vado a trovare.

Un pensiero speciale è rivolto ai miei meravigliosi amici Alessandro, Beatrice, Edoardo e Leonardo: siete stati in tutti questi anni parte fondamentale della mia felicità e delle mie risate. Con ciascuno di voi conservo ricordi indimenticabili e storie da raccontare, e posso considerarvi a pieno titolo parte della mia famiglia. Siete tra le persone che mi conoscono più a fondo e per le quali nutro il massimo rispetto. Senza di voi questo percorso sarebbe stato meno luminoso e meno vivo. Spero che il legame che ci unisce non si perda mai.

Un pensiero felice va anche ai tanti amici incontrati durante il primo e il secondo semestre di Erasmus a Vienna. L'Erasmus è qualcosa di speciale: in pochi mesi si creano legami intensi e profondissimi, capaci di lasciare un segno che dura ben oltre il tempo trascorso insieme. Con voi ho vissuto esperienze emozionanti e momenti di pura spensieratezza, tra risate, viaggi e scoperte che hanno reso questa avventura indimenticabile. Grazie a ciascuno di voi, italiani e internazionali, per aver reso questi dieci mesi probabilmente i più belli della mia vita, un periodo che custodirò nel cuore e ricorderò per sempre. In particolare, un grazie speciale ad Antonio, che nei primi mesi della mia esperienza è stato un compagno insostituibile: sempre presente e disponibile,

un amico eccezionale.

Desidero inoltre ringraziare i due relatori, il Prof. Matteo Maria Pelagatti e il Prof. Paolo Maranzano, che hanno seguito la mia tesi con costanza e attenzione durante questi mesi. Mi hanno fatto sentire sempre supportato, dimostrando un autentico interesse per il progetto e accompagnandomi con suggerimenti preziosi e indicazioni concrete su come procedere. La loro disponibilità, unita all'approccio proattivo e partecipe, ha reso questo lavoro non solo più solido, ma anche più stimolante e arricchente. La loro dedizione all'insegnamento e alla ricerca è stata per me un esempio e una fonte di ispirazione.

Un grazie sincero va anche a tutti coloro che hanno incrociato il mio cammino e che non ho citato. In particolare, ai professori che ho avuto il piacere di conoscere in Italia e all'Estero e ai compagni di università, con cui ho condiviso lezioni, studio e momenti preziosi che hanno arricchito il mio percorso.

Infine, desidero profondamente ringraziare la persona a cui dedico questa tesi, Beatrice. Sei stata la sorpresa più bella di questo ultimo anno e, oltre a rendermi una persona migliore, mi rendi felice. Sono onorato e orgoglioso di essere accanto a te e non vedo l'ora di affrontare le sfide che la vita ci offrirà insieme. Vorrei che l'impegno e la dedizione che ho messo in questo lavoro possano essere lo specchio dell'amore e della cura che ogni giorno provo per te. Questa tesi non è solo un traguardo personale, ma anche un passo del nostro cammino comune: a te dedico ogni sforzo, ogni pagina, ogni traguardo.

Abstract

This thesis studies the aggregation of political opinion polls in Italy (2018–2024), focusing on improving the accuracy and interpretability of “poll of polls” estimates. Existing approaches often neglect systematic differences across institutes and rarely disentangle variance from bias in polling errors. To address these issues, I develop a family of state-space models that extend the Pelagatti–Monti framework by incorporating institute-specific house effects, heteroskedastic variances linked to sample size, binomial measurement structures, and a covariance matrix capturing structural dependencies among institutes. Estimation relies on Kalman filtering, Durbin–Koopman approximations and importance sampling.

Results show that variance is the main driver of discrepancies, while institute-specific biases are generally modest (around ± 1 percentage point) and party-specific. A few institutes display persistent deviations, whereas others differ mainly because of higher sampling variance. Structural correlations suggest small clusters of methodological affinity rather than broad polarization. Smoothed vote-share trends align with well-known electoral dynamics. Overall, the thesis demonstrates that effective poll aggregation requires bias correction, variance-based weighting, and structural diagnostics, offering a replicable methodology with broader applications beyond the Italian case.

Declaration of Generative AI Use

This master's thesis employed Generative Artificial Intelligence (GenAI) tools in a limited and controlled manner, in compliance with the Recommendations for the Use of Generative Artificial Intelligence in the Drafting of Student Papers (29 March 2024), issued by the “Digital Transformation and Wellbeing LAB” and the Local Node of the CINI National Laboratory on Informatics and Society, in collaboration with the Department of Informatics, Systems and Communication of the University of Milano-Bicocca.

GenAI was used only for idea generation, content structuring, and linguistic refinement, never as a direct source of factual information. All outputs were critically reviewed, adapted, and verified to prevent hallucinations, outdated content, or plagiarism, and to ensure alignment with the author's own reasoning. Sources mentioned in AI-generated content were independently checked for accuracy. The interaction followed a semi-structured, author-controlled approach, with the final responsibility for all arguments and conclusions resting entirely with the author.

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Introduction

In the landscape of modern political analysis, polls of polls have become central. These are methods that aggregate polls from multiple survey organizations to produce estimates that are less sensitive to the quirks of any single poll. Because media-commissioned polls often use samples too small to detect the relatively modest day-to-day or week-to-week movements in voter sentiment, aggregation is typically necessary to obtain stable, daily estimates (Jackman, 2005). The widespread availability of poll numbers online has further lowered the cost of aggregation and helped popularize election forecasting.

A concise historical account is offered by Blumenthal (2014), who traces the evolution from simple averages to the debate between descriptive “measurement” and predictive “forecast” approaches. The first high-profile media uses date to 1992: The Economist published a weekly average of about six national “trial heat” polls, and soon after William Schneider at CNN adopted the same idea on air to compare results across organizations and mitigate single-poll errors (Blumenthal, 2014).

Aggregators operate using mechanisms ranging from simple rolling averages to the calculation of weighted averages based on criteria such as sample size, recency, and the institution’s past reliability. A more sophisticated version is that of FiveThirtyEight: Nate Silver weights each poll based on the pollster’s historical track record, sample size, and recentness of the poll, considering also demographic influences (Hillygus, 2011).

As Pasek (2015) notes, the main advantage is perspective: averaging multiple polls usually yields a more reliable and comprehensive picture of a race than relying on any single result. However, there are important caveats. Some aggregators include all polls regardless of methodological quality; if low-quality pollsters field many surveys, their results can dominate the aggregate. Moreover, aggregators are only as good as the underlying polls: if many pollsters share a systematic error in the same direction, pooling can propagate (or even amplify) that error. Another related concern is herd-

ing, the tendency of pollsters to converge toward one another late in the campaign, potentially reducing apparent dispersion without eliminating bias.

Finally, the growing popularity of aggregation has raised worries that media organizations might invest more in aggregators or modelers and less in conducting high-quality, probability-based surveys, surveys that are costly and labor-intensive. If the stock of high-quality data shrinks, aggregate estimates may become less informative.

When combining results from different pollsters, it is common to observe systematic differences between their estimates, even when surveys are conducted at the same time. These differences, known as house effects, reflect persistent biases linked to a pollster’s methodology, variations in mode of interview (telephone, face-to-face, or Internet), sampling and weighting procedures, the day of the week or the time of day a company interviews, the age, ethnic and gender composition of a given company’s interviewers, question wording and question ordering (Jackman, 2005). For example, one firm may consistently report slightly higher support for a given party than others, not because public opinion has actually shifted, but because of the way that firm conducts its polls.

Alongside bias, another important dimension is the polling variance of each organization. This represents how much a pollster’s results tend to fluctuate from one survey to another due to sampling error and other sources of random noise. Two pollsters might have similar house effects (or none at all) but very different levels of variance: one producing relatively stable results, the other showing large swings.

Building on this background, the empirical analysis in this study starts from the “poll of polls” framework proposed by Professor Matteo Pelagatti and Professor Gianna Monti. In its basic form, their model treats the true level of support for a party as a latent quantity evolving as a random walk process over time, observed through noisy measurements provided by multiple polling organizations. The approach assumes a common trend for all pollsters, no systematic differences between them (no house effects), and a constant measurement variance across organizations. This specification already allows for smoothing short-term fluctuations and producing a coherent time

series of voter support estimates from multiple surveys day by day.

In the present work, I extend this baseline in three directions that directly address the issues outlined above. First, I allow for the estimation of house effects, enabling the model to detect and quantify persistent differences between pollsters that remain even after accounting for sampling noise. Second, I relax the assumption of constant measurement variance by letting each polling organization have its own variance parameter, capturing differences in precision across survey houses. Third, I introduce a structural covariance specification in which pollster deviations are not modeled as fixed mean shifts but as correlated innovations. This extension captures persistent methodological affinities or common influences between institutes, providing a network view of inter-pollster dependencies.

In addition to these extensions, I explicitly consider two statistical formulations for the target variable. A Gaussian specification (in line with Pelagatti and Monti) treats poll percentages as continuous and approximately normal, while a binomial specification models the actual count of respondents favoring a party out of the survey sample size. The latter approach allows the model to incorporate the discrete nature of polling data and the direct link between variance and sample size, while the Gaussian approach offers computational simplicity and interpretability.

These enhancements make it possible not only to adjust the aggregated trend for systematic biases and differing levels of reliability, but also to decompose each pollster's mean squared error (MSE) into its two main components: squared bias and variance. In addition, the structural covariance specification goes beyond the identification of house effects by highlighting methodological clusters and interdependencies between polling institutes. Such a decomposition and network perspective provide a clearer picture of the origins of polling errors and allow for more informed weighting and interpretation of survey data.

The remainder of this thesis is organized as follows. Chapter 1 reviews the relevant literature, beginning with a historical overview of political polling and the emergence of poll aggregation, followed by a previous modeling approaches. Chap-

ter 2 presents the methodological framework, describing the data sources used in the analysis and detailing the statistical techniques implemented, including both Gaussian and binomial state-space formulations with extensions for estimating house effects, pollster-specific variances, the structural cross-institute covariance specification, and the decomposition of mean squared error. Chapter 3 reports the empirical results obtained by applying these models to the dataset, highlighting differences between pollsters and comparing the performance of the various specifications. Finally, Chapter 4 concludes, summarizing the main contributions and outlining possible directions for future research.

1 Literature Revision

1.1 History of Polls

Since the advent of modern scientific polling in the 1936 U.S. presidential election, public opinion surveys have become a central feature of democratic politics, particularly in the United States. As Hillygus (2011) notes, “Elections and polling are so intertwined that it is hard to imagine one without the other”. Initially developed to forecast election outcomes, polls have expanded in purpose to understanding voter behavior and informing campaign strategy. Over the past decades, changes in technology, media, and statistical techniques have heavily shaped the way polls are conducted, interpreted, and disseminated. In parallel, the rise of poll aggregation websites and alternative forecasting methods has sparked debates about the purpose, accuracy, and future of election polling (Blumenthal, 2014).

1.1.1 Origins of Polls

The roots of election polling go back at least to the 1824 presidential election when informal “trial heats” were recorded in taverns and public meetings (Smith, 1990). The turning point came in 1936, when George Gallup’s quota-controlled survey outperformed the massive but biased straw poll conducted by *The Literary Digest* (Crossley, 1937). This success marked the birth of modern scientific polling, based on representative sampling, and established a new standard for measuring public opinion.

By the 1960s and 1970s, skepticism about the predictive value of small-sample surveys had largely given way to confidence in probability sampling (Converse, 2009). Polling became a fixture of campaigns, not only for prediction but also for strategic planning and issue testing. However, the industry has always been subject to periodic crises of credibility, from the notorious 1948 “Dewey Defeats Truman” miscalculation (when the *Chicago Daily Tribune* and others prematurely declared “Dewey Defeats Truman,” only for Truman to win in a stunning election upset) to more recent debates over

Internet-based techniques.

Technological innovation has dramatically increased the volume of election polling. Traugott (2005) estimated a nine-fold increase in “trial heat” polls between 1984 and 2000, a growth trend that accelerated with the introduction of interactive voice response (IVR) and online polling after 2000 (Hillygus, 2011). In the 2008 election alone, over 975 presidential trial heat questions were asked, representing well over a million interviews between Labor Day and Election Day (Panagopoulos, 2009).

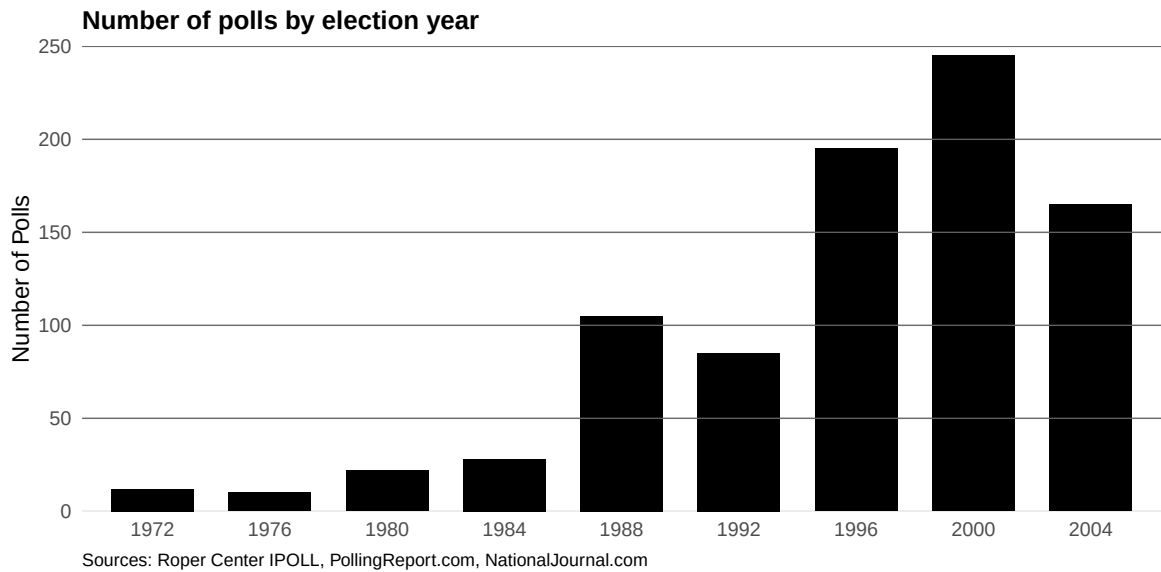


Figure 1: Increase in “trial heat” polls between 1984 and 2000 in USA.

Alongside volume growth, the nature of polling has shifted. Once dominated by live-interviewer telephone surveys commissioned by media outlets or campaigns, today’s polling includes self-sponsored polls by entrepreneurial pollsters, online opt-in panels, and automated IVR surveys. The dissemination channels have also changed: whereas journalists once acted as “gatekeepers” of polling quality, results are now published directly on blogs, social media, and dedicated aggregation sites, reducing editorial filtering (Blumenthal, 2014) (Blumenthal, 2005).

1.1.2 Polls Aggregation

The practice of aggregating polls, combining results from multiple surveys to produce a single indicator, has a relatively recent history in electoral politics, but its conceptual

roots can be traced back to early attempts by journalists and analysts to reduce the noise inherent in individual polls. According to Blumenthal (2014), the first notable instance in a major news outlet occurred in 1992, when *The Economist* published a weekly “poll of polls” during the U.S. presidential campaign. This average, based on approximately half a dozen national trial-heat results from surveys conducted by other media organizations, aimed to contextualize individual poll findings and mitigate the impact of sampling error. In the same year, William Schneider, CNN’s senior political analyst, introduced the “poll of polls” to American television audiences. After reporting the latest CNN–USA Today–Gallup poll, Schneider explained that he also looked at an average of other organizations’ surveys because “all such surveys are subject to error” and he wanted to “see how much consistency there is across all of the polls” (Blumenthal, 2014). He further noted that all six surveys he examined produced results “within the margin of error of the averages we’re looking at right now.”

While initially a supplementary feature, CNN’s own poll at the time showed President Clinton with a smaller lead than others, the aggregation concept quickly became a regular element of CNN’s election coverage by the 1996 cycle. This early adoption underscores one of aggregation’s enduring purposes: to place individual poll results in broader context and assess their convergence or divergence from a general trend.

The method’s fame expanded significantly with the advent of the Internet. In 2000, the then-nascent RealClearPolitics.com took the aggregation model further by not only producing national averages similar to Schneider’s, but also compiling and averaging polls from 21 battleground states (Blumenthal, 2014)). Unlike CNN, RealClearPolitics included results from automated, recorded-voice surveys polls (known as interactive voice response or IVR), broadening the scope of data sources.

The 2004 election cycle marked an inflection point. The explosive growth of online political discourse and the accessibility of poll data fostered the emergence of multiple aggregation platforms. In 2006, Charles Franklin and Mark Blumenthal launched Pollster.com, which innovated beyond simple averages by offering interactive,

graphical displays of poll trends. These developments aligned with what Hillygus (2011) observed as a broader technological transformation in polling dissemination: poll results were increasingly reported directly on blogs and aggregation sites rather than filtered through traditional media “gatekeepers”.

Among the various poll aggregation platforms that emerged in the mid-2000s, FiveThirtyEight.com, launched by Nate Silver in March 2008, quickly distinguished itself through a blend of transparency, methodological sophistication, and probabilistic forecasting. Silver built credibility by accurately projecting the outcomes of key primary contests, including the 2008 “Super Tuesday” primaries. Unlike simple averaging approaches that treat all polls equally, FiveThirtyEight’s model incorporates multiple weighting factors: the recency of the poll (with an exponential decay formula privileging fresh data), the sample size (adjusting for the margin of error), and the historical accuracy of the polling organization, with adjustments for systematic bias (what Silver calls pollster-introduced error or PIE). This weighting scheme aims to counteract random sampling error and recurrent methodological skew.

One of the model’s distinctive features is its trendline adjustment, which uses national polling data to update estimates in under-pollled states, allowing results from demographically similar states to influence each other. States are grouped according to variables in political history, religious composition, ethnic and racial demographics, economic structure, and age and education profiles. This “demographic similarity” approach enables the model to produce estimates even where polling is sparse.

FiveThirtyEight’s methodology also includes Monte Carlo simulations, running 10,000 election scenarios to capture the probability distribution of possible outcomes. This probabilistic framing (expressed as a percentage chance of victory for each candidate) was a key innovation in communicating forecast uncertainty to the public, contrasting sharply with deterministic “calls” used by media outlets.

As Felder (2009) has observed, this probabilistic framing “shifted the conversation from who will win to how likely each outcome is,” introducing a more nuanced understanding of electoral uncertainty into mainstream political coverage. By empha-

sizing probabilities over certainties, Silver’s model encouraged both journalists and the public to interpret polls not as definitive predictions but as evolving indicators within a broader range of possible scenarios. This shift represented, in Felder’s words (2009), “a change in the language of political forecasting” that better reflected the inherent variability of electoral dynamics.

Despite its sophistication, the model is not immune to challenges. As documented in the 2008 cycle, trendline adjustments proved sensitive to short-term “bounces” following events such as party conventions or economic shocks. Nevertheless, the 2008 performance validated the model’s general approach: on Election Day, FiveThirtyEight projected a 98.9% probability of an Obama win and forecasted 52.3 % of the popular vote (actual: 52.9%).

By combining rigorous weighting, demographic modeling, and simulation, Silver’s platform helped shift public and journalistic expectations about polling aggregation from static “poll of polls” averages to dynamic, probability-based electoral forecasts. Its success also underscored a broader trend identified by Hillygus (2011) and Blumenthal (2014): the integration of statistical modeling with vast, freely accessible polling datasets, transforming the consumption and interpretation of election polling in the digital era.

Despite the advances brought by poll aggregation and probabilistic models, some concerns were raised within the polling profession. Gallup editor-in-chief Frank Newport speculated after 2012 that if organizations focused on aggregating and analyzing others’ polls rather than conducting their own, “fewer and fewer polls [would be] left to aggregate and put into statistical models” (Newport, 2012). This “law of the commons” worry, however, is tempered by the continued availability of inexpensive automated polls and the fact that aggregation serves a distinct function from original polling, particularly where many surveys ask identical questions with comparable methodology.

Another major concern about polls aggregation regards methodological standpoint, Hillygus (2011) cautions that naïvely averaging polls assumes that individual

“house effects” will cancel out. If biases are systematic (such as telephone polls underrepresenting cell-phone-only households) aggregation could amplify rather than reduce error. Sophisticated models like FiveThirtyEight attempt to mitigate these issues through weighting schemes and selective inclusion criteria.

A further complication is the phenomenon of “convergence” or “herding,” in which polls tend to align more closely as Election Day nears, whether because voter preferences stabilize or because pollsters make methodological adjustments to avoid being perceived as outliers (Linzer, 2012). This dynamic has been amplified in the era of poll aggregation and probabilistic forecasting, where models like FiveThirtyEight explicitly translate polling data into win probabilities, reinforcing the association between polls and electoral prediction.

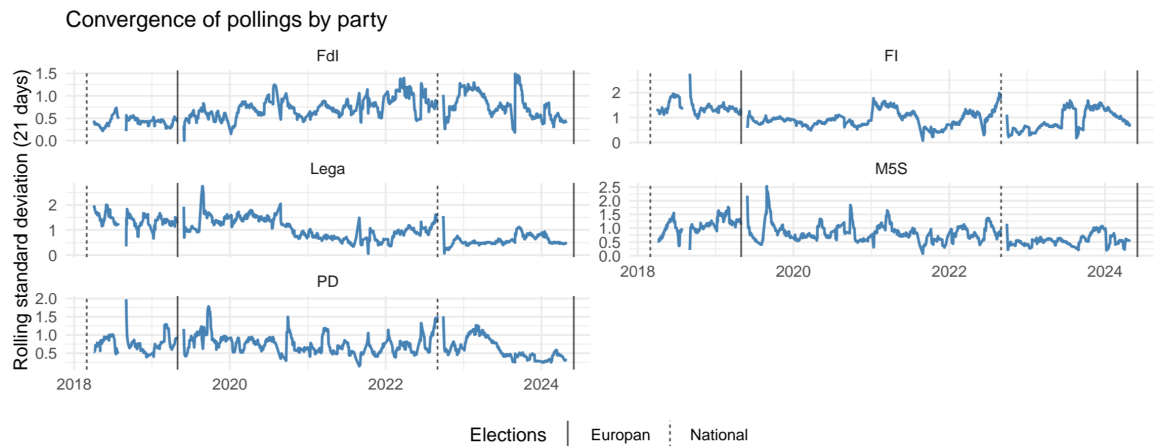


Figure 2: Herding. Notes: values are hidden for the 21 days following each election (vertical lines).

In sum, poll aggregation has evolved from a journalistic experiment into a central component of contemporary election forecasting, driven by the growth of polling volume, the democratization of data access, and the integration of advanced statistical modeling. This transformation has redefined how both the media and the public engage with polling, embedding probabilistic reasoning into political discourse and altering the language of electoral prediction.

1.1.3 Alternative forecasting approaches

A persistent tension in the interpretation of election polls lies between their role as a “snapshot in time” and their perceived function as forecasts. Pollsters typically stress the former, noting that results can shift as campaigns evolve. Yet, as Blumenthal (2014) observes, public engagement with polls peaks in the days immediately preceding an election, precisely when their perceived predictive value is highest.

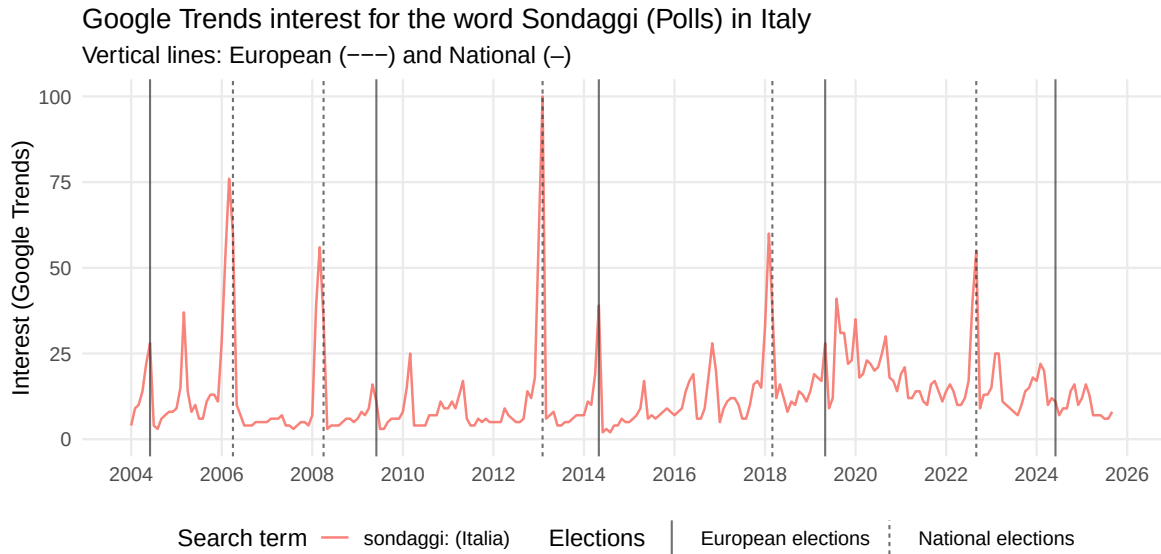


Figure 3: Public interest with polls peaks in the days preceding an election.

Even if traditional opinion polls remain the primary instrument for measuring electoral dynamics, alternative forecasting approaches have gained ground. Macroeconomic models, drawing on variables such as GDP growth, unemployment and inflation, generate theory-driven predictions (for example Abramowitz’s model 2004).

Moreover prediction markets bring together the bets placed by participants who are presumed to have relevant knowledge about the election, using the resulting prices as a collective indicator of expected outcomes (Berg et al., 2003). These methods can sometimes outperform polls early in the campaign cycle but often rely on polling data as an input (Hillygus, 2011).

Campaigns themselves have increasingly integrated polling with large-scale voter databases, combining registration files with consumer and demographic data to

create granular profiles. As Hillygus and Shields (2014) document, this practice enables campaigns to deliver highly tailored messages, often on polarizing issues, to specific voter segments, thus diminishing the role of public polling as the sole instrument for strategic decision-making.

The evolution of election polling mirrors broader transformations in media, technology, and political strategy. From Gallup’s pioneering quota samples to Silver’s probabilistic aggregation models, the field has grown in both volume and methodological sophistication. As Hillygus (2011) notes, polls are now increasingly “supplemented or replaced by alternate measures and methods”, yet they remain central to electoral strategy, media narratives, and public discourse. The integration of advanced modeling, as exemplified by FiveThirtyEight, has shifted expectations from static “snapshots” toward dynamic probability estimates, embedding statistical reasoning into how modern democracies interpret electoral competition.

1.2 Introduction to State-Space Models

In this study, as well as in a large body of research on electoral polling, one of the most frequently employed methodological approaches is the state-space model, a statistical framework that allows researchers to capture the dynamic evolution of public opinion over time while accounting for both observed data and underlying latent processes.

State-space models provide a flexible and powerful framework for time-series analysis. The key idea is to view the observed series y_t as arising from the interaction of unobserved components, collectively called the state vector α_t , that may represent features such as trend, seasonality, regression effects, or the impact of shocks (Durbin and Koopman, 2001).

As stated by Pelagatti’s book (2015), linear Gaussian state-space model is defined by two equations:

Observation equation:

$$y_t = Z_t \alpha_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, H_t). \quad (1)$$

State (transition) equation

$$\alpha_{t+1} = T_t \alpha_t + R_t \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q_t). \quad (2)$$

Here

- y_t is the vector of observations at time t ;
- α_t is the (unobserved) state vector at time t ;
- Z_t links the state to the observation;
- T_t governs the law of motion of the state;
- R_t maps the state disturbances η_t into the state space;
- H_t and Q_t are covariance matrices for observation and state disturbances, respectively.

The stochastic assumptions on the disturbance sequences are as follows:

- $\{\varepsilon_t\}$ is a sequence of serially uncorrelated random vectors with

$$\mathbb{E}[\varepsilon_t] = 0, \quad \text{Var}[\varepsilon_t] = H_t;$$

- $\{\eta_t\}$ is a sequence of serially uncorrelated random vectors with

$$\mathbb{E}[\eta_t] = 0, \quad \text{Var}[\eta_t] = Q_t;$$

- ε_t and η_s are uncorrelated for all t, s ;
- the initial state α_1 is specified by

$$\mathbb{E}[\alpha_1] = a_{1|0}, \quad \text{Var}[\alpha_1] = P_{1|0}.$$

If α_1 , ε_t and η_t are normally distributed, the system is referred to as a *Gaussian state-space model*.

The main inferential task is to form statistical predictions of the state vector α_t given the information set $\mathcal{Y}_s = \{y_1, \dots, y_s\}$. The three classical problems are:

- *Forecasting*: prediction of α_t based on $\mathcal{Y}_s$ with $s < t$;
- *Filtering*: prediction of α_t based on $\mathcal{Y}_t$;
- *Smoothing*: prediction of α_t based on $\mathcal{Y}_s$ with $s > t$.

We denote the linear prediction of the state vector given $\mathcal{Y}_s$ by

$$a_{t|s} = \mathbb{E}[\alpha_t \mid \mathcal{Y}_s], \quad P_{t|s} = \text{Var}[\alpha_t \mid \mathcal{Y}_s],$$

where $P_{t|s}$ is the conditional variance-covariance matrix of α_t .

For linear-Gaussian models, inference revolves around estimating the current state α_t and its uncertainty as each new observation arrives. This is accomplished efficiently by the Kalman filter, a recursive algorithm that alternates between:

1) *Prediction (a priori step)*. From $(a_{t-1|t-1}, P_{t-1|t-1})$, predict the next state and its covariance

$$a_{t|t-1} = T_{t-1}a_{t-1|t-1}, \tag{3}$$

$$P_{t|t-1} = T_{t-1}P_{t-1|t-1}T_{t-1}^\top + R_{t-1}Q_{t-1}R_{t-1}^\top. \tag{4}$$

2) *Update (a posteriori step)*. When y_t is observed, compute the innovation $v_t = y_t - Z_t a_{t|t-1}$, its variance $F_t = Z_t P_{t|t-1} Z_t^\top + H_t$, the Kalman gain $K_t = P_{t|t-1} Z_t^\top F_t^{-1}$, and update

$$a_{t|t} = a_{t|t-1} + K_t v_t, \tag{5}$$

$$P_{t|t} = P_{t|t-1} - K_t F_t K_t^\top. \tag{6}$$

Intuitively, the filter blends prior information (the prediction from the state equation) with the new data point, weighting them by their respective uncertainties.

The state-space framework also accommodates non-Gaussian observations by choosing an observational distribution from the exponential family and linking its mean to the latent state via a suitable link function. Let the linear predictor be

$$\theta_t = Z_t \alpha_t, \quad l(\mu_t) = \theta_t,$$

where $l(\cdot)$ is the link and $\mu_t = \mathbb{E}(y_t \mid \theta_t)$. Common choices include Poisson, binomial, negative binomial, and gamma observations.

Example (binomial state-space).

$$y_t \sim \text{Bin}(n_t, \pi_t), \quad \theta_t = \text{logit}(\pi_t), \quad \alpha_{t+1} = T_t \alpha_t + R_t \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q_t).$$

In this case the observation density is non-Gaussian, and the link between the state α_t and the observations is non-linear. For this reason the classical Kalman filter cannot be applied directly. A common way forward is to replace the true non-Gaussian model with a locally Gaussian approximation, so that we can still use Kalman filtering and smoothing.

One standard approach is the Laplace approximation, implemented via an iteratively reweighted least squares (IRLS) algorithm (Helske, 2017). The idea is to approximate the contribution of each observation y_t by a quadratic expansion of its log-likelihood $\log p(y_t \mid \theta_t)$ around the current estimate of $\theta_t = Z_t \alpha_t$. This expansion defines pseudo-observations $\tilde{y}_t$ with a corresponding Gaussian error variance H_t , leading to a local model of the form

$$\tilde{y}_t = Z_t \alpha_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, H_t).$$

The algorithm then proceeds iteratively:

1. Compute $\tilde{y}_t$ and H_t from the current estimate of θ_t ;
2. Apply Kalman filtering and smoothing on the Gaussian approximation to update the estimate of the states;
3. Repeat until the estimates converge.

At convergence, the algorithm returns the posterior mode of $p(\theta \mid y)$ for the original non-Gaussian model. In other words, the Kalman filter is used inside an iterative scheme that gradually adapts the Gaussian approximation until it matches the non-Gaussian likelihood.

Because the inverse link function $l^{-1}(\cdot)$ is non-linear, transforming the mode $\hat{\theta}_t$ can give biased estimates of $\mu_t = \mathbb{E}(y_t \mid \theta_t)$. To correct for this and to obtain likelihood values and prediction intervals, one typically uses *importance sampling*. The idea is to simulate state trajectories from the approximating Gaussian model and then reweight them by the ratio of the true non-Gaussian likelihood to the Gaussian pseudo-likelihood. This produces unbiased estimates of expectations such as $\mathbb{E}[f(\alpha_t) \mid y]$ (e.g. $f(\alpha_t) = l^{-1}(Z_t \alpha_t)$), as well as approximations to the log-likelihood and valid prediction intervals.

1.3 Pre-existing Models

As already stated, a well-established approach to combining election polls is to treat the “true” level of support for a party as an unobserved quantity that evolves smoothly over time, while each poll is a noisy, and potentially biased, measurement of that latent process. This idea is central to Simon Jackman’s influential pool-of-polls state-space model (Jackman, 2005), which provides a formal statistical framework for integrating information across survey organisations and over time.

In Jackman’s formulation, the observation from poll i , conducted by organisation j at time t , is written as:

$$p_i \sim \mathcal{N}(m_i, \sigma_i^2)$$

with the mean structure:

$$m_i = a_t + \delta_j$$

where:

- a_t is the latent, bias-free level of party support at time t ,
- δ_j is the house effect for pollster j , capturing systematic over- or underestimation,
- σ_i^2 is the sampling variance of poll i , usually computed from the reported proportion p_i and sample size n_i as

$$\sigma_i^2 \approx \frac{p_i(1 - p_i)}{n_i}.$$

The latent process a_t is modelled as a random walk:

$$a_t = a_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \nu^2).$$

which encodes the belief that voter sentiment changes incrementally rather than jumping erratically from day to day. This temporal smoothing reduces the influence of short-lived fluctuations caused by sampling noise.

Identification of the δ_j parameters requires an additional constraint. Jackman anchors the model by including the final election outcome as a “poll” with zero measurement error on election day, which pins down the overall level of a_t and thus allows estimation of each pollster’s bias relative to the truth.

The model can be expressed in standard state-space form, with observation equation:

$$p_i = Z_i \alpha_t + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, H_i),$$

where Z_i is a design vector selecting the relevant states (a_t and the δ_j for pollster j) and $H_i = \sigma_i^2$ is the measurement variance. State equation:

$$\alpha_t = T \alpha_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q),$$

with $\alpha_t = (a_t, \delta_1, \dots, \delta_J)^\top$ transition matrix T that keeps biases constant over time and Q containing the process variance ν^2 for a_t and zeros for the δ_j .

Estimation is performed in a Bayesian framework using Markov chain Monte Carlo (MCMC) sampling. This yields posterior distributions for the latent support a_t , the house effects δ_j and the process variance ν^2 .

In his application to the 2004 Australian federal election, Jackman found that much of the variation between simultaneous polls could be attributed to persistent house effects rather than genuine shifts in opinion. Correcting for these biases produced a much cleaner, more stable trajectory of public support. An important practical implication is that, with typical sample sizes of 1,000–1,500 respondents, day-to-day changes of one percentage point are often indistinguishable from noise. Without pooling across pollsters and over time, most “movements” in the polls are likely artefacts of sampling error.

In summary, Jackman’s pool-of-polls model remains a cornerstone of modern poll aggregation: it explicitly models a smooth latent trend, corrects for systematic biases, and weights polls according to their precision. These same principles (bias adjustment, variance-based weighting, and temporal smoothing) underpin many later developments, including the more flexible formulations applied in the present study.

An alternative approach, adopted in several studies by Erikson and Wlezien (1999), combines polls conducted by multiple organizations during the election campaign with the aim of analyzing the evolution of public opinion. As in Jackman’s case, the importance of correcting for house effects is explicitly recognized. The basic structure of the model is as follows:

$$v_i = \gamma_{t_i} + \delta_{j_i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$

where:

- v_i is the vote share estimated by poll i for the party of interest;
- γ_{t_i} represents the true level of electoral support on day t_i when the poll was

conducted;

- δ_{j_i} is the systematic bias of polling organization j that conducted the poll;
- ε_i is a random error with variance σ^2 , capturing sampling noise.

The model is implemented as a fixed-effects regression (*analysis of covariance*), in which:

- each polling organization, except one, is represented by a dummy variable (the omitted organization serves as the reference category to identify the δ_j);
- each campaign day is also represented by a dummy variable, thus providing separate estimates of vote share for each date, already adjusted for systematic differences between pollsters.

In this framework, the daily coefficients (γ_t) provide a time series of vote intentions purged of house effects, while the coefficients associated with pollster dummies (δ_j) give a direct estimate of each pollster’s bias relative to the reference organization.

Compared with Jackman’s model, this approach is statistically simpler: it does not impose a temporal dependence structure (such as a random walk on the latent trend) and does not require Bayesian estimation. Nevertheless, thanks to its linear fixed-effects formulation, it allows immediate control for systematic differences between pollsters and a clean reconstruction of the trend in vote intentions over time.

A significant contribution in this line of research is offered by Fisher et al. (2011), who propose a state-space model for estimating vote intentions in Britain. The basic framework is analogous to Jackman’s: poll observations are treated as noisy and biased measures of a latent process a_t , which represents the true level of electoral support on each day of the campaign. However, the model introduces some important methodological extensions.

Formally, for poll i conducted by organization j at time t , the assumption is:

$$p_i \sim \mathcal{N}(m_i, \sigma_i^2), \quad m_i = a_t + \delta_j.$$

As in Jackman, a_t is the latent vote intention and δ_j represents the house effect, the systematic tendency of each polling organization to over- or underestimate party support.

The novelty lies in the treatment of sampling variance. Fisher et al. (2011) do not assume that sampling variance is exactly given by the binomial formula $p_i(1-p_i)/n_i$. Instead, they introduce a multiplier s^2 , defined as the design effect, which captures systematic deviations due to weighting practices, survey mode (for example online panels), or other sources of non-random error:

$$\sigma_i^2 = s^2 \cdot \frac{p_i(1-p_i)}{n_i}.$$

This correction acknowledges that real-world polls often exhibit larger (or smaller) errors than would be expected under simple random sampling.

Vote intention a_t follows a random walk with variance τ_1^2 :

$$a_t \sim \mathcal{N}(a_{t-1} + h_t, \tau_1^2),$$

where a drift term h_t is introduced, itself modeled as a further random walk with variance τ_2^2 . This specification allows for the representation of medium-term trends beyond purely daily fluctuations.

Another crucial element is the treatment of the identification problem for the δ_j . Similarly to what Jackman did, Fisher et al. (2011) recognize the possibility of an average bias across the entire polling industry (industry bias). Anchoring is achieved by using the previous election result, treated as an observation without error. In this way, the series a_t is pinned to a known real value at one observable point, allowing estimation of each pollster's relative bias without assuming that their average cancels out.

A more recent contribution to the literature is offered by Professor Pelagatti and Monti (2024), who developed a state-space framework designed to aggregate a large panel of Italian polls collected since 2018. Their goal is to construct a coherent daily

estimate of voting intentions from data that are asynchronous, irregular in time, and heterogeneous in sample size. The central idea is to treat the true daily voting intention for a party, denoted μ_t , as a latent process observed through noisy measurements provided by multiple polling organizations. For institute i on day t , the reported proportion is written as

$$y_{i,t} = \mu_t + \varepsilon_{i,t},$$

with zero-mean error $\varepsilon_{i,t}$ and variance inversely proportional to the sample size:

$$\text{Var}(\varepsilon_{i,t}) = \frac{\sigma_\varepsilon^2}{n_{i,t}}.$$

This assumption implies that all pollsters are unbiased on average and equally reliable once differences in sample size are taken into account. Unlike other models, no explicit correction for systematic house effects is introduced at this stage. Missing data, which arise naturally since not all institutes publish polls every day, are handled seamlessly by the Kalman filter. The temporal evolution of voting intentions is described by a random walk:

$$\mu_{t+1} = \mu_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2),$$

which embodies the belief that public opinion moves gradually rather than erratically. The variance parameter σ_η^2 controls the degree of smoothness: small values lead to a stable trajectory, while larger values allow for faster shifts in support.

In matrix notation (with Y_t the $I \times 1$ vector of observed vote shares on day t , where I is the number of institutes considered):

Observation equation:

$$Y_t = Z\alpha_t + \varepsilon_t, \quad Z = \mathbf{1}_{I \times 1}, \quad \varepsilon_t \sim \mathcal{N}(0, H_t),$$

with

$$H_t = \text{diag}\left(\frac{\sigma_\varepsilon^2}{n_{1,t}}, \dots, \frac{\sigma_\varepsilon^2}{n_{I,t}}\right).$$

State equation:

$$\alpha_t = \mu_t, \quad \alpha_{t+1} = T\alpha_t + \eta_t, \quad T = 1, \quad \eta_t \sim \mathcal{N}(0, Q),$$

where

$$Q = \sigma_\eta^2.$$

Since μ_t is a percentage constrained to a plausible range (roughly between 5% and 35% for major parties), an informative prior is specified at the initial time.

Estimation is performed in a frequentist framework using maximum likelihood and the Kalman filter as implemented in the R package **KFAS** (Helske, 2017). Both the innovation variance of the random walk and the scale of the measurement error are estimated from the data, while the smoother produces daily estimates of party support along with confidence intervals. On days without polls, the filter naturally interpolates the missing information, with greater uncertainty reflecting the absence of direct observations.

Conceptually, this model shares with Jackman’s pool-of-polls the central notion of a latent trajectory smoothed through time. However, it differs in three key respects: it assumes no systematic pollster bias, it weights observations purely through their declared sample sizes and it relies on a frequentist rather than Bayesian estimation strategy. Compared with the fixed-effects regression approach of Erikson and Wlezien, the Pelagatti-Monti specification does not estimate separate parameters for each pollster or campaign day. Instead, it uses the state-space structure to interpolate and smooth across time. Relative to the more elaborate formulation of Fisher and collaborators, it is more parsimonious, foregoing both the explicit treatment of industry-wide bias and the modelling of medium-term drifts.

The main strength of the Pelagatti-Monti model lies in its balance of simplicity and realism. It provides an effective way to pool asynchronous data from dozens of institutes, automatically accounts for heteroscedastic measurement error due to varying sample sizes and yields stable daily trajectories that can be readily compared across

parties. At the same time, the framework is flexible: extensions could easily incorporate house effects, different weighting mechanism (variance structure) or additional drift terms if required. In its baseline form, however, it already offers a transparent and efficient method for transforming fragmented polling data into coherent time series of voting intentions.

2 Methodology

2.1 Data

The dataset employed in this study consists of an extensive collection of opinion polls conducted in Italy over a period of more than six years. In total, it includes over 12,000 individual observations, each representing the outcome of a single poll on a political party at a specific point in time. Each row in the dataset corresponds to the result of a party in a given survey and contains five key variables (as shown in Table 1):

1. the name of the political party,
2. the date when the poll was published,
3. the percentage of support reported in that poll,
4. the name of the polling institute responsible for the survey,
5. the sample size used for the estimation.

This structure allows for a systematic and consistent analysis of party support over time, while retaining information about methodological factors such as the polling institute and the size of the respondent pool, which may influence the reliability of the reported results.

The time span of the dataset ranges from 14 March 2018 to 30 April 2024. This period is particularly significant for Italian politics, as it begins immediately after the 2018 general elections and ends roughly six weeks before the 2024 European elections. The six-year horizon captures a politically turbulent era, during which Italy experienced considerable governmental and institutional instability.

In fact, throughout this timeframe the country witnessed the succession of four different governments: the first Conte government (Conte I), the second Conte government (Conte II), the technical executive led by Mario Draghi, and finally the Meloni government. Each of these governments was marked by different coalitions and

Party	Date	Value	Institute	Sample Size
FdI	14/03/2018	3.1	SWG	1000
FI	14/03/2018	10.5	SWG	1000
Lega	14/03/2018	22.3	SWG	1000
M5S	14/03/2018	34.5	SWG	1000
PD	14/03/2018	18	SWG	1000
FdI	15/03/2018	4.0	Piepoli	512
FdI	15/03/2018	4.8	Demos & Pi	1246
FI	15/03/2018	14.0	Piepoli	512
FI	15/03/2018	12.8	Demos & Pi	1246
Lega	15/03/2018	18.0	Piepoli	512
Lega	15/03/2018	18.2	Demos & Pi	1246
M5S	15/03/2018	34.0	Piepoli	512
M5S	15/03/2018	33.8	Demos & Pi	1246
PD	15/03/2018	18.5	Piepoli	512
PD	15/03/2018	18.4	Demos & Pi	1246
FdI	19/03/2018	4.2	Euromedia	800
FdI	19/03/2018	4.0	Piepoli	508
FI	19/03/2018	13.4	Euromedia	800
FI	19/03/2018	13.5	Piepoli	508
...	...	...	...	...

Table 1: Example of polling data included in the dataset.

power balances, which inevitably shaped the trajectory of public opinion and voting intentions.

The dataset includes results for a broad range of political parties, though the primary focus is on the mainstream actors that dominate the Italian political arena. These are:

- Lega (Lega),
- Partito Democratico (PD),
- Fratelli d’Italia (FdI),
- Movimento 5 Stelle (M5S),
- Forza Italia (FI),
- Azione (Az),
- Italia Viva (IV).

For the purposes of the present analysis, two parties, Azione (Az) and Italia Viva (IV), were excluded. The reason is that their historical presence in the dataset is inconsistent: initially, they appear as separate entities, and later they are sometimes reported jointly due to alliances or joint parliamentary groups. This inconsistency would complicate the construction of a coherent time series and risk introducing artificial discontinuities in the analysis. Their exclusion ensures the comparability and homogeneity of the parties under investigation.

The dataset brings together surveys from a wide variety of polling agencies, with more than 40 different institutes represented. However, for reasons of computational efficiency and to reduce analytical complexity, the attention in this study has been narrowed to the 16 institutes with the largest number of surveys conducted. This choice ensures both the robustness of the analysis and the representativeness of the overall dataset: together, these selected institutes account for the vast majority of all recorded polls. The 16 most active and influential institutes are as shown in Table 2:

Institute	N
Tecnè	1817
SWG	1693
Termometro Politico	1353
EMG	868
Euromedia	841
Piepoli	642
Noto Sondaggi	570
Ixè	553
Lab2101	510
Ipsos	508
Quorum	448
Demopolis	437
BiDiMedia	413
Index	407
Index Research	336
Demos & Pi	249
EMG Acqua	175
Eumetra	173
Scenari Politici-Winpoll	127
Winpoll	48
...	...

Table 2: Top 20 polling institutes by number of polls. The 16 most frequent are highlighted in bold.

Each of these agencies has its own methodological approaches, sampling strategies, and historical reputations, which can influence both the accuracy and the comparability of their results. For example, some institutes consistently report larger or smaller figures for specific parties compared to others, a phenomenon sometimes referred to as “house effects” in polling research. By focusing on the most prolific agencies, it becomes possible to control for these effects while maintaining a sufficiently large sample size for meaningful statistical analysis.

A central feature of the dataset is the sample size reported for each poll. Sample size is a critical parameter because it directly influences the reliability and precision of survey estimates: larger samples generally reduce the margin of error, whereas smaller samples may produce less stable results. An inspection of the data reveals significant variability across institutes as shown in Table 3:

Institute	Mean	Median	Min	Max	Std. Dev.	% Null	Total
Tecnè	1095.557	1000.000	791	16000	866.075	1.541	1817
SWG	1290.732	1200.000	1000	2100	167.832	0.000	1693
Termometro Politico	3221.323	3200.000	1000	6000	824.941	0.000	1353
EMG	1528.976	1526.000	1000	2000	184.438	4.378	868
Euromedia	876.973	800.000	800	1159	102.801	4.518	841
Piepoli	524.218	505.000	356	1000	97.525	49.221	642
Noto Sondaggi	1035.897	1000.000	1000	2000	186.513	65.789	570
Ixè	1001.950	1000.000	1000	1209	20.110	3.074	553
Lab2101	1036.721	1000.000	587	4100	340.204	2.353	510
Ipsos	1005.063	1000.000	600	1503	78.068	2.953	508
Quorum	789.319	802.000	500	1602	173.005	4.688	448
Demopolis	1785.434	1500.000	1000	8860	941.894	8.696	437
BiDiMedia	1742.422	1455.000	769	12000	1425.426	1.211	413
Index	808.235	800.000	800	1200	56.884	16.462	407
Index Research	811.147	800.000	800	1500	87.766	6.548	336
Demos & Pi	1041.494	1007.000	917	1522	108.166	0.000	249

Table 3: Summary statistics for the sample size variable of the main polling institutes (values rounded to three decimals).

An in-depth inspection of Table 3, which reports descriptive statistics for sample sizes across the main polling institutes, highlights the considerable heterogeneity in methodological practices.

At first glance, it is possible to observe that most institutes converge around

a “standard” sample size of approximately one thousand respondents. This figure is widely recognized in survey research as a threshold that balances representativeness and cost-effectiveness, making it the default choice for many polling organizations. Institutes such as Ixè, Ipsos, Lab2101, Noto Sondaggi, and Demos & Pi adhere closely to this standard, with both mean and median values consistently around one thousand units.

Nevertheless, significant heterogeneity emerges when analyzing the data in more detail. Some institutes clearly distinguish themselves by adopting considerably larger samples. Termometro Politico, for instance, reports an average sample size exceeding 3,200 units, with a median very close to that value, which demonstrates a systematic preference for larger-scale surveys. Similarly, Demopolis and BiDiMedia also stand out for their use of larger-than-average samples (around 1,700 respondents). The methodological implication is that surveys conducted by these institutes potentially benefit from greater statistical precision, especially in estimating proportions within the population, though at the cost of increased fieldwork and resource demands.

In contrast, other institutes operate with significantly smaller samples. Piepoli, Index, and Index Research typically rely on samples ranging from 500 to 800 respondents, with corresponding median values confirming this tendency. Smaller sample sizes inevitably imply higher margins of error, which can limit the precision of estimates and make fluctuations across time or across surveys more difficult to interpret. This difference in methodological approach between institutes may therefore partly explain divergences in reported results, especially when margins of error are not explicitly communicated.

The variability of sample sizes is another relevant dimension. Standard deviation values highlight how some institutes show a high degree of consistency, while others present remarkable fluctuations. Ixè and Ipsos, for example, have very limited variability (below 100), indicating a highly standardized and stable sample design across their surveys. This methodological rigor facilitates comparability across time and suggests a deliberate strategy to maintain methodological homogeneity. Conversely, BiDiMedia

and Demopolis are characterized by exceptionally high variability, with standard deviations of over 1,400 and 900 respectively. This reflects the fact that while these institutes sometimes employ sample sizes close to the standard, in other instances they conduct surveys on exceptionally large samples (up to 12,000 respondents for BiDiMedia and over 8,800 for Demopolis). Although these outliers may improve the robustness of specific surveys, they reduce overall comparability and complicate longitudinal analyses, since sample design changes dramatically between surveys.

The range of minimum and maximum values confirms this interpretation. Institutes such as Tecné, BiDiMedia, and Demopolis display extreme ranges: Tecné, for instance, reports sample sizes spanning from fewer than 800 respondents to as many as 16,000. These wide fluctuations suggest a flexible or opportunistic approach to sample design, possibly linked to the nature of the survey or the resources available at the time. In contrast, other institutes show far greater stability: Ixè maintains a minimum of 1,000 and a maximum of 1,209 respondents, while Ipsos ranges from 600 to 1,503. Such stability contributes to methodological transparency and enhances comparability over time.

Another dimension worth considering is the presence of missing data on sample sizes. The percentage of null values is not negligible in some cases and raises concerns regarding transparency and completeness of information. Noto Sondaggi, for instance, records missing values in almost two-thirds of its surveys (65.8%), while Piepoli lacks information for nearly half of its sample (49.2%). This strongly limits the possibility of reconstructing methodological patterns and undermines the reliability of comparative analyses. Other institutes, such as SWG and Demos & Pi, report no missing values, thereby offering a more consistent and transparent dataset.

In summary, although most Italian polling institutes rely on a conventional sample size of around 1,000 respondents, the evidence shows significant differences in both the magnitude and stability of samples across organizations. However, it is important to stress that sample size alone cannot be considered a definitive indicator of survey quality. The reliability of estimates also depends on other methodological

dimensions, such as the sampling frame, weighting procedures, response rates, and fieldwork techniques. For example, a survey with 500 well-stratified respondents may produce more accurate results than one with several thousand respondents selected through less rigorous methods. Moreover, differences in methodological choices are often reflected in the so-called house effects, that is, systematic deviations in survey results that can be attributed to the specific practices, assumptions, or adjustments applied by each institute. These effects highlight that comparisons across polls must consider not only the nominal size of the sample but also the broader methodological context in which surveys are conducted.

Finally a consistent color palette was defined for the major political parties, assigning easily recognizable colors (in particular green for Lega, red for PD, black for FdI, golden yellow for M5S, blue for FI). This step, while not strictly necessary for the dataset itself, is important for the clarity and interpretability of subsequent visualizations.

In addition to the Italian corpus, a comparable dataset was also collected for Austria, obtained from the Europe Elects project (Elects, 2025). This dataset contains approximately 2,400 observations, spanning the period from December 2017 to March 2025, and it is structured in the same way as the Italian one: each row represents the result of a single party in a specific poll and includes the party name, the publication date, the reported percentage, the polling institute, and the sample size. The dataset covers the main Austrian political parties, such as the ÖVP, SPÖ, FPÖ, the Greens, and NEOS, and therefore provides a similarly comprehensive picture of public opinion dynamics.

Although fully comparable in terms of structure and methodology, the Austrian dataset is not analyzed systematically in this thesis. The focus remains on Italy, given the broader scope, larger number of observations, and the political relevance of the Italian case study. Nevertheless, all the methodological procedures and analytical steps applied to the Italian data can, and indeed have been, replicated on the Austrian dataset. For reasons of space and focus, however, the results of these replications are

reported briefly only in the Appendix of this work, but the availability of a parallel dataset and the parallel application of the techniques underscore the robustness and transferability of the analytical framework employed.

2.2 Analysis Techniques

Exactly like in the Pelagatti and Monti model, the basic idea is that all polls measure the same latent level (trend) μ_t (expressed in percentage points) but that each polling institute exhibits a systematic shift δ_j (the so-called house effect). Moreover, polls do not all have the same precision: larger samples are, on average, more precise.

We consider three state-space specifications: (i) a Gaussian model with institute bias and institute-specific variances; (ii) a Binomial state-space model; (iii) a Binomial state-space model with institute bias; (iv) a Gaussian model with structural cross-institute dependence.

Indices are institutes $j = 1, \dots, p$ and days $t = 1, \dots, T$. For a fixed party, $y_{j,t} \in [0, 100]$ denotes the reported percentage, $n_{j,t} \in \mathbb{N}$ the sample size, μ_t the latent trend, and δ_j the institute-specific bias. A sum-to-zero constraint $\sum_{j=1}^p \delta_j = 0$ ensures identifiability.

2.2.1 Gaussian model with bias and institute-specific variances

We can therefore formulate everything in a heteroskedastic Gaussian model:

$$y_{j,t} = \mu_t + \delta_j + \varepsilon_{j,t}, \quad \varepsilon_{j,t} \sim \mathcal{N}\left(0, \frac{\sigma_j^2}{n_{j,t}}\right),$$

where $y_{j,t}$ is the observed support for a party (in percentage points) in the poll t conducted by the institute j , μ_t is the latent trend, δ_j is the house effect, and the variance depends on the sample size $n_{j,t}$.

The latent trend evolves over time according to a local-level model:

$$\mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, q).$$

It is important to note that this formulation assumes Gaussian errors. In principle, a binomial (or multinomial) model would be more appropriate since poll outcomes are based on counts of respondents. However, the Gaussian approximation is often used in practice because it simplifies estimation and is reasonable when sample sizes are large and proportions are not too close to the limits 0 or 1.

Considering the compact form of the model, we stack the observations as

$$y_t = (y_{1,t}, \dots, y_{p,t})^\top \in \mathbb{R}^p$$

and define the state vector as

$$\alpha_t = (\mu_t, \delta_1, \dots, \delta_{p-1})^\top \in \mathbb{R}^p.$$

To identify the model, we impose a sum-to-zero constraint on the house effects,

$$\sum_{j=1}^p \delta_j = 0 \quad \Rightarrow \quad \delta_p = -\sum_{j=1}^{p-1} \delta_j,$$

so that only $p - 1$ bias components are free in the state. This leads to the classical state-space system:

$$y_t = Z\alpha_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, H_t),$$

$$\alpha_t = T\alpha_{t-1} + R\eta_t, \quad \eta_t \sim \mathcal{N}(0, Q),$$

with

$$T = I_p, \quad R = (1, 0, \dots, 0)^\top \in \mathbb{R}^{p \times 1}, \quad Q = q \in \mathbb{R}^{1 \times 1},$$

$$H_t = \text{diag}\left(\frac{\sigma_1^2}{n_{1,t}}, \dots, \frac{\sigma_p^2}{n_{p,t}}\right),$$

and

$$Z = \begin{bmatrix} 1 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & 1 \\ 1 & -1 & -1 & \cdots & -1 \end{bmatrix},$$

where the first column loads μ_t on all institutes and the last row enforces $\delta_p = -(\delta_1 + \cdots + \delta_{p-1})$.

We use weakly informative initial conditions: a very large variance for μ_1 (so that the early dynamics are driven by the data), and moderately dispersed priors centered at zero for the δ_j 's.

Given a set of parameters, filtering proceeds with the standard Gaussian recursions.

Prediction:

$$a_{t|t-1} = T a_{t-1|t-1}, \quad P_{t|t-1} = T P_{t-1|t-1} T^\top + R Q R^\top.$$

Innovation and its variance:

$$v_t = y_t - Z a_{t|t-1}, \quad F_t = Z P_{t|t-1} Z^\top + H_t.$$

Update:

$$K_t = P_{t|t-1} Z^\top F_t^{-1}, \quad a_{t|t} = a_{t|t-1} + K_t v_t, \quad P_{t|t} = P_{t|t-1} - K_t F_t K_t^\top.$$

At each t , the prediction provides the best estimate before seeing the new polls; the innovation v_t is the surprise (observed minus expected); and the gain K_t determines how much to trust this surprise relative to the current uncertainty ($P_{t|t-1}$) and the measurement noise (H_t). If a poll is more precise (large $n_{j,t}$, small σ_j^2), it will receive more weight in the update.

We collect the parameters as

$$\theta = (\log q, \log \sigma_1^2, \dots, \log \sigma_p^2).$$

Working on the log scale ensures positivity and improves numerical stability. For a given θ , the Kalman filter produces innovations v_t and variances F_t , from which the Gaussian log-likelihood is obtained as:

$$\ell(\theta) = -\frac{1}{2} \sum_{t=1}^T \left\{ \log |F_t| + v_t^\top F_t^{-1} v_t + p \log(2\pi) \right\}.$$

Operational recipe.

1. **Log-parametrization.** Work in the log-space, so that q and σ_j^2 are guaranteed to be positive through the exponential transform.
2. **Evaluate $\ell(\theta)$ via the filter.** For each candidate θ , run the filter through the full series and accumulate the log-likelihood contributions ℓ_t .
3. **Maximize $\ell(\theta)$.** Use a numerical optimizer (for example BFGS or L-BFGS) with reasonable initial values. If the optimizer minimizes the *negative* log-likelihood (as in `optim` in R), the observed Hessian at the optimum already corresponds to the information matrix.
4. **Standard errors.** At the optimum $\hat{\theta}$, approximate the covariance matrix as the inverse of the observed Hessian. By maximum-likelihood theory, the estimator $\hat{\theta}$ is asymptotically normal with covariance given by the inverse of the Fisher information matrix. In practice we use the observed information at the optimum (the inverse of the Hessian of the minimized objective function).
5. **Back-transform to the natural scale.** Report $\hat{\sigma}_j^2 = \exp(\hat{\theta}_j)$. If SE_j is the standard error of $\hat{\theta}_j$, a 95% confidence interval for σ_j^2 on the *natural scale* is obtained by exponentiating the bounds:

$$95\% \text{ CI: } [\exp(\hat{\theta}_j - 1.96 SE_j), \exp(\hat{\theta}_j + 1.96 SE_j)].$$

The same applies to q .

The filter summarizes all the information in the data into the innovations v_t and their variances F_t . Parameters that make the surprises small (and/or their expected variances large) maximize the likelihood. The log parametrization prevents the optimizer from searching in non-admissible regions.

After completing the filter, we refine the estimates at all time points using backward smoothing. Given $a_{T|T}, P_{T|T}$, the recursions are

$$\begin{aligned} J_t &= P_{t|t} T^\top P_{t+1|t}^{-1}, & a_{t|T} &= a_{t|t} + J_t(a_{t+1|T} - a_{t+1|t}), \\ P_{t|T} &= P_{t|t} + J_t(P_{t+1|T} - P_{t+1|t})J_t^\top, & t &= T-1, \dots, 1. \end{aligned}$$

The trend estimate is given by

$$\hat{\mu}_t = a_{t|T}^{(1)}, \quad SE(\hat{\mu}_t) = \sqrt{P_{t|T}(1, 1)}.$$

The house effects are static by construction (they do not evolve over time). Therefore, the smoothed estimates are constant in t ; one can read them from $a_{t|T}$ at any time point (by convention we report $t = T$). For $j = 1, \dots, p-1$,

$$\hat{\delta}_j = a_{T|T}^{(j+1)}, \quad SE(\hat{\delta}_j) = \sqrt{P_{T|T}(j+1, j+1)}.$$

For δ_p we use the identification constraint:

$$\hat{\delta}_p = - \sum_{j=1}^{p-1} \hat{\delta}_j, \quad \text{Var}(\hat{\delta}_p) = \mathbf{1}^\top \Sigma_{\delta\delta} \mathbf{1},$$

where $\Sigma_{\delta\delta}$ is the $(p-1) \times (p-1)$ covariance matrix of $(\delta_1, \dots, \delta_{p-1})^\top$ and $\mathbf{1} = (1, \dots, 1)^\top$.

The corresponding 95% confidence interval is

$$\hat{\delta}_p \pm 1.96 \sqrt{\mathbf{1}^\top \Sigma_{\delta\delta} \mathbf{1}}.$$

By construction, $\hat{\delta}_j > 0$ indicates that institute j systematically reports values higher than the cross-sectional average, while $\hat{\delta}_j < 0$ implies the opposite. These shifts are measured in percentage points and represent persistent biases relative to the latent trend, not random fluctuations. Hence, the house effects capture structural differences across pollsters, summarizing how each institute's results deviate from the common underlying trend.

Relative to the bias-free trend, the mean squared error at the institute level can be heuristically decomposed as

$$MSE_j \equiv \mathbb{E}_t[(y_{j,t} - \mu_t)^2] \approx \delta_j^2 + \mathbb{E}_t\left[\frac{\sigma_j^2}{n_{j,t}}\right],$$

that is, the sum of a systematic error (δ_j^2) and a sampling/measurement variance. In practice, the second term is summarized by $\sigma_j^2/\bar{n}_j$, where $\bar{n}_j$ is the average sample size for institute j over the period. All quantities are expressed in percentage points, so both $\hat{\mu}_t$ and $\hat{\delta}_j$ can be directly interpreted on that scale.

This decomposition should not be understood as the exact MSE_j . First, the sampling variance varies over time with $n_{j,t}$, so replacing it with an average is only an approximation. Second, the latent trend μ_t is unobserved and must itself be estimated. Still, the formula is useful as an interpretive device: if δ_j^2 dominates, differences across pollsters reflect persistent structural biases (house effects); if instead the sampling variance term is larger, discrepancies are mainly due to random noise driven by finite sample sizes. Hence, the comparison between the two components highlights whether institute-level errors are primarily systematic or stochastic.

2.2.2 Binomial state-space model (bias-free)

Here polls are treated as counts:

$$x_{j,t} \mid \pi_t \sim \text{Binomial}(n_{j,t}, \pi_t), \quad y_{j,t} = 100 x_{j,t}/n_{j,t}.$$

All institutes measure the same latent probability π_t , linked to the state by

$$\pi_t = \text{logit}^{-1}(\mu_t), \quad \mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, q).$$

This yields a non-Gaussian state-space model. Exact likelihood is unavailable; we use the classical Gaussian approximation of Durbin and Koopman (2001), an iterative re-weighting on the logit scale. For each candidate q , pseudo-observations are built, the Kalman filter/smoothing applied, and the approximate log-likelihood maximized over $\log q$.

We use a weakly informative proper prior on the state,

$$\mu_1 \sim \mathcal{N}(a_1, P_1),$$

with $a_1 \approx -1.4$ ($\text{logit}^{-1}(a_1) \approx 0.20$, i.e. an initial support around 20%) and $P_1 = 0.3$. This allows the early dynamics to be driven largely by the data while keeping the filter numerically stable.

Posterior trend estimates are obtained from smoothed states. On the probability scale,

$$\hat{\pi}_t = \text{logit}^{-1}(a_{t|T}), \quad \hat{y}_t = 100 \hat{\pi}_t.$$

Because the Gaussian approximation may understate uncertainty, especially near boundaries, we refine by *importance sampling*: draw trajectories from the Gaussian smoother, reweight by the true binomial likelihood, and compute weighted expectations/quantiles. This delivers unbiased estimates of π_t and proper credible intervals.

2.2.3 Binomial state-space model with institute bias

This specification merges the two ideas already introduced: binomial observations and institute-specific house effects. For a fixed party, let $x_{j,t} \in \{0, \dots, n_{j,t}\}$ be the number of respondents in poll t by institute j who declare support (the published percentage

is $y_{j,t} = 100 x_{j,t}/n_{j,t}$). Conditional on a common trend and an institute shift,

$$x_{j,t} \mid \mu_t, \delta_j \sim \text{Binomial}(n_{j,t}, \pi_{j,t}), \quad \pi_{j,t} = \text{logit}^{-1}(\mu_t + \delta_j),$$

where the latent level μ_t evolves as a local level and the biases are static:

$$\mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, q), \quad \delta_j = \text{const. in } t.$$

For identification we impose $\sum_{j=1}^p \delta_j = 0$.

Stacking states $\alpha_t = (\mu_t, \delta_1, \dots, \delta_{p-1})^\top$, the transition equation is

$$\alpha_t = T\alpha_{t-1} + R\eta_t, \quad T = I_p, \quad R = (1, 0, \dots, 0)^\top, \quad \eta_t \sim \mathcal{N}(0, q).$$

Estimation proceeds exactly as in the bias-free binomial model via iterative Gaussian approximation, with maximization over $\log q$. Uncertainty bands are again refined by importance sampling.

House effects are read from the static components of the smoothed state, with $\hat{\delta}_j = a_{T|T}^{(j+1)}$ for $j = 1, \dots, p-1$ and $\hat{\delta}_p = -\sum_{j=1}^{p-1} \hat{\delta}_j$. Since δ_j acts on log-odds, its impact in percentage points depends on the evaluation level. For a chosen reference probability π^* (on election day),

$$\text{Bias}_j(\pi^*) = 100 \left\{ \text{logit}^{-1}(\text{logit}(\pi^*) + \delta_j) - \pi^* \right\},$$

In practice we use π^* at a specific calendar date to rank institutes and to produce credible intervals for the bias in percentage points via weighted quantiles of the sampled δ_j 's mapped through the transformation above. Compared to the Gaussian bias model, measurement variance is now implied by the binomial law, $\pi_{j,t}(1 - \pi_{j,t})/n_{j,t}$, so institute-specific variances need not be estimated.

To validate the estimated house effects, we compare them with the *realized*

bias observed at the 2022 election. For each institute j , let

$$\text{Bias}_j^{\text{real}} = y_{j,\text{pre}}^{\text{last}} - y^{\text{vote}},$$

where $y_{j,\text{pre}}^{\text{last}}$ is the last poll published before election day and y^{vote} is the official election result (percentage points). This difference provides an ex-post measure of bias, directly comparable with the model-based bias $\text{Bias}_j^{\text{model}}(\pi^*)$ obtained by mapping δ_j on the percentage scale at the election reference level π^* .

We summarize agreement between the two notions through graphical inspection, plotting $\text{Bias}_j^{\text{model}}$ against $\text{Bias}_j^{\text{real}}$ with the 45° identity line. The comparison shows whether house effects extracted from the full time series anticipate the systematic deviations that appear when institutes are judged against the realized outcome. High concordance indicates that the state-space model successfully disentangles persistent institute-level shifts from random sampling noise.

2.2.4 Gaussian model with structural cross-institute dependence

In continuity with the previous models, we assume that all surveys measure the same latent level (trend) μ_t in percentage points, which evolves according to a local level specification. The novelty here is that we do not model house effects as fixed level shifts (δ_j); instead, we allow the deviations around μ_t to be correlated across institutes through a structural, time-invariant covariance matrix. In this way, we capture methodological affinities or common influences without imposing constant mean shifts for each institute.

For institutes ($j = 1, \dots, p$) and days ($t = 1, \dots, T$), let $y_{j,t} \in [0, 100]$ denote the reported percentage and $n_{j,t}$ the sample size. We set

$$y_{j,t} = \mu_t + \varepsilon_{j,t}, \quad \boldsymbol{\varepsilon}_t \equiv (\varepsilon_{1,t}, \dots, \varepsilon_{p,t})^\top \sim \mathcal{N}(0, H_t),$$

$$\mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, q).$$

The observational error component is the sum of two parts:

$$H_t = \underbrace{\Sigma_{\text{bias}}}_{\text{structural, constant in } t} + \underbrace{\text{diag}\left(\frac{\sigma_1^2}{n_{1,t}}, \dots, \frac{\sigma_p^2}{n_{p,t}}\right)}_{\text{sampling, heteroskedastic}}.$$

Σ_{bias} is a $(p \times p)$ positive-definite matrix that captures persistent co-movements across institutes; the diagonal term decreases with the sample size and represents sampling noise (institute-specific via σ_j^2).

In compact form:

$$y_t = \mathbf{1}_p \mu_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, H_t), \quad \mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, q).$$

Compared to the Gaussian model with bias (δ_j), here the mean shift is not identified at the institute level: the conditional mean is μ_t , while possible systematic 'house effects' manifest as covariances among errors, rather than as separate intercepts.

To guarantee positivity and numerical stability, we work with:

$$q = \exp(\theta_1), \quad \sigma_j^2 = \exp(\theta_{1+j}), \quad j = 1, \dots, p.$$

For Σ_{bias} we adopt a Cholesky factorization with exponentiated diagonal:

$$\Sigma_{\text{bias}} = LL^\top, \quad L = \begin{bmatrix} e^{\ell_{11}} & 0 & \dots & 0 \\ \ell_{21} & e^{\ell_{22}} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{p1} & \ell_{p2} & \dots & e^{\ell_{pp}} \end{bmatrix}.$$

The free parameters are $\theta = (\log q, \log \sigma_1^2, \dots, \log \sigma_p^2, \text{vec}(L)_{\text{lower-triangular}})$. Given θ , H_t varies over time via $n_{j,t}$, whereas Σ_{bias} remains constant. The Gaussian likelihood is obtained through the standard Kalman filter for a linear-Gaussian model with time-varying H_t ; maximization is performed by ML.

Besides $\hat{q}$ and $\hat{\sigma}_j^2$, the main object of inference is

$$\widehat{\Sigma}_{\text{bias}} \quad \text{and the corresponding} \quad \widehat{\text{Corr}}_{\text{bias}} = \text{cov2cor}(\widehat{\Sigma}_{\text{bias}}).$$

The entries $\widehat{\text{Corr}}_{\text{bias},jk}$ measure the extent to which the (structural) deviations of institutes j and k move together after removing the common trend and correcting for sampling noise. High values suggest methodological affinities or shared sources/procedures; values close to zero indicate independence; negative values indicate systematic movements in opposite directions.

Comparing this model with the previous ones we can highlight:

- **vs. Gaussian with bias (δ_j).** In that case, cross-institute differences were modeled as *fixed levels* (δ_j) with independent variances. Here, instead, we do not impose mean shifts and allow *dependencies* to emerge through Σ_{bias} . This is particularly useful when survey houses share shocks or practices that generate co-movements, rather than constant mean offsets.
- **vs. binomial models (with/without bias).** We remain within a Gaussianized framework, with heteroskedasticity ($\propto 1/n_{j,t}$) estimated via σ_j^2 , instead of being derived from the binomial law. We gain computational simplicity and the ability to estimate a full Σ_{bias} , but give up exact consistency with binomial counts (acceptable for moderate-to-large samples and proportions away from the boundaries).

3 Results

In this section I present empirical evidence on Italian national polling for 2018–2024. I begin with the *party-level* analysis, comparing institutes along a common set of diagnostics (house effects, institute-specific variance, and MSE). I then turn to the *institute-level* analysis, which summarises cross-party regularities (bias heatmap, comparative panels, and aggregate rankings with and without within-party standardisation).

3.1 Party-level Analysis

The empirical results are organised by party. For each of the five major parties (Lega, PD, FdI, M5S, FI) I apply the three specifications introduced in Chapter 2: the Gaussian model with institute bias and institute specific variances (`gauss_bias_var_mod`), the Binomial state-space model without bias (`bin_mod`), and the Binomial model with institute bias (`bin_bias_mod`). This party-by-party layout makes it easy to compare pollster behaviour across parties while keeping the interpretation within each party self contained.

For each party I discuss, in order, the following families of plots:

1. Estimated institute bias (house effect) with 95% CI. Ranking of polling organisations by the estimated house effect. Positive values mean systematic overestimation of the party relative to the common trend; negative values indicate underestimation. Intervals crossing zero suggest no statistically detectable bias. Unless otherwise noted, biases are reported in percentage points; when estimated on the logit scale (binomial model), they are mapped into percentage points at a reference level (the party share on election day) as described in Section 2.3.
2. Institute-specific variances 95% CI (Gaussian model). Comparison of the estimated measurement variances σ_j^2 for each institute (reported in squared percentage points and scaled by the model), which capture the typical random fluctuation of a poll from that organisation after accounting for sample size. Smaller values

indicate more stable measurements.

3. MSE composition by institute. Bar chart that decomposes the mean squared error into its two components, *bias*² and *variance* (see Section 2.3.1). This allows us to see whether the typical error of an institute is driven mainly by a persistent shift (house effect) or by sampling/measurement noise.
4. Estimated share over time with 95% credible interval (Binomial model). Smoothed time series of party support obtained from the binomial state-space model; credible bands are computed via importance sampling to correct the Gaussian approximation. This figure is the poll-of-polls backbone and is used as the reference trend when discussing institute behaviour.
5. Actual vs estimated bias (validation). Scatterplot comparing, for each institute, the modelled bias and the realised bias observed at the 2022 election (last pre-election poll minus official result). Points near the 45° line and more in general a positive correlation indicate good agreement between the inferred house effects and the ex-post deviations.

Throughout the chapter I keep axis units in percentage points for levels and biases, use consistent institute ordering within each panel and highlight noteworthy outliers only when they materially affect interpretation. The same template is repeated for each party so that cross-party comparisons are straightforward; party colours follow the palette defined in the Data section.

3.1.1 Lega

Lega (Lega per Salvini Premier) is a right-wing populist Italian party led by Matteo Salvini. Founded in 1991 as the Northern League from a federation of northern regional movements, it later broadened nationwide. It advocates stricter immigration controls, federalism, and a Eurosceptic stance, and often partners in centre-right coalitions. In the European Parliament, its MEPs align with the Patriots for Europe group (formerly

Identity & Democracy). The results for Lega highlight both persistent differences across polling institutes and substantial shifts in public support during the 2018-2024 period.

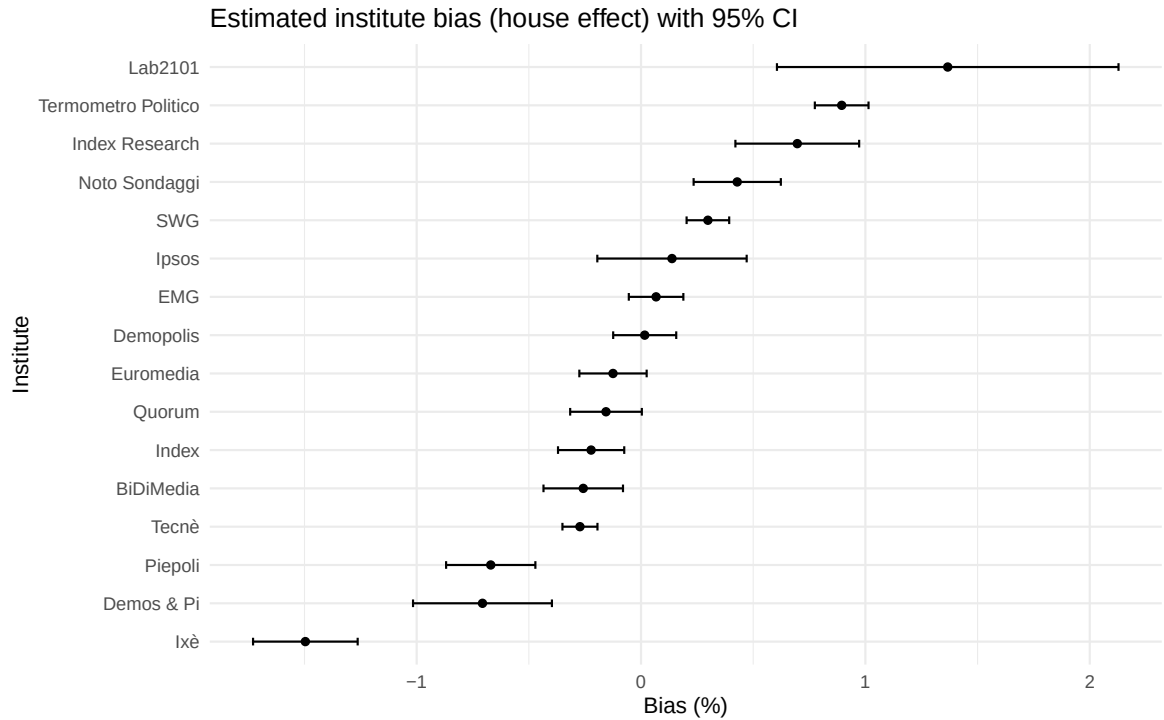


Figure 4: House effects for Lega (point estimate and 95% CI).

Figure 4 shows that most institutes cluster near zero, with meaningful departures for a few. Lab2101 and Termometro Politico tend to report systematically higher levels of support for Lega, while Ixè, Demos & Pi, and Piepoli usually understate the party’s strength. These deviations are generally within a couple of percentage points but are persistent over time, which is consistent with methodological heterogeneity across organisations (for example, different modes, distinct weighting/post-stratification schemes, or question wording/order). Substantively, such house effects can nudge media narratives by giving Lega slightly more or less visibility depending on the commissioning outlet.

Figure 5 highlights notable dispersion in precision across pollsters. Some institutes deliver relatively stable measures (lower variance), whereas others show wider fluctuations around the latent trend (Ipsos and Termometro Politico). Part of this dispersion reflects design choices: larger or more consistently stratified samples reduce

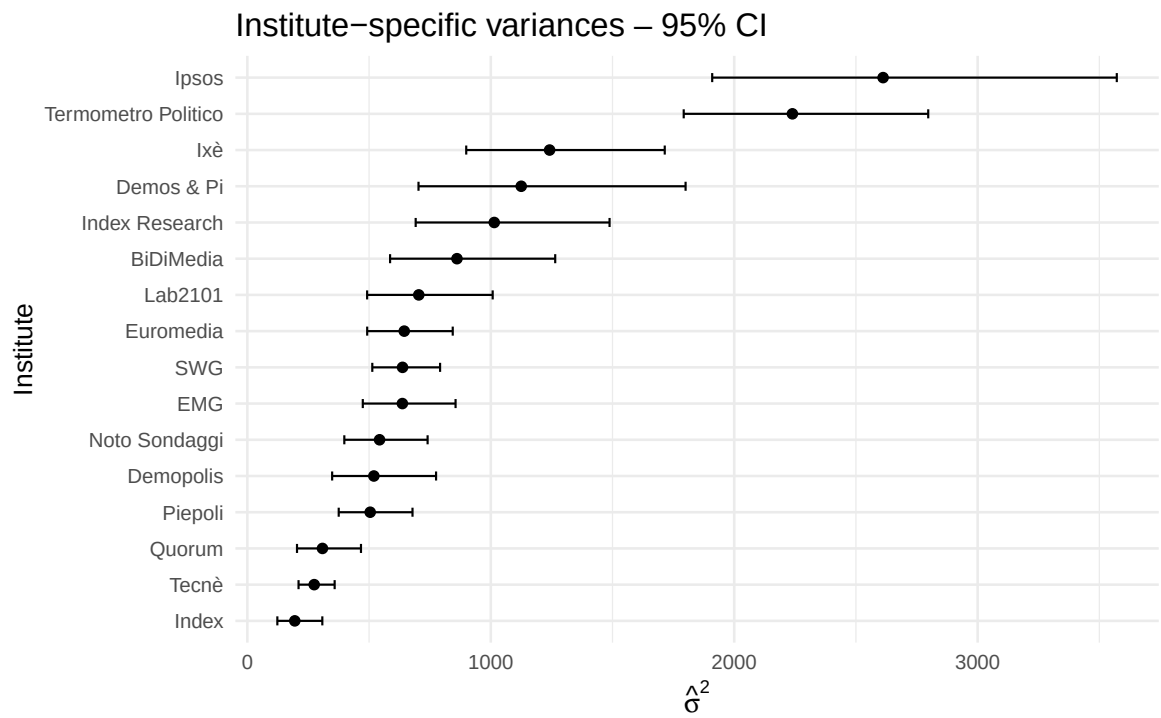


Figure 5: Estimated institute-specific variances (95% CI) under the Gaussian model.

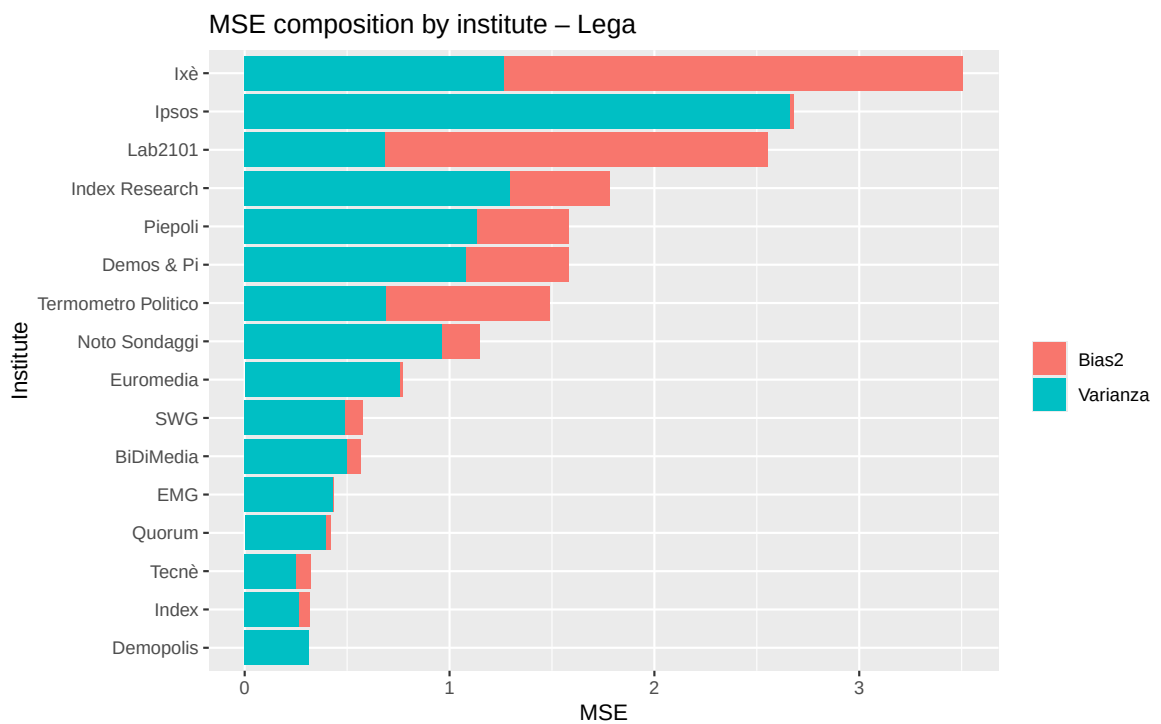


Figure 6: MSE decomposition into $bias^2$ and variance for Lega, by institute.

variance; smaller or irregular samples increase it. For Lega, whose support displays marked medium-run movements, higher measurement variance tends to exaggerate short-term swings, potentially amplifying perceived volatility in the public debate.

In Figure 6, the mean squared error (MSE) is decomposed into variance and squared bias. For most institutes, the variance component dominates, indicating that cross-pollster disagreements mostly stem from sampling/measurement noise rather than from structural miscalibration. However there is a notable share of institutes with larger house effects (Lab2101, Ixè, Termometro Politico), the squared-bias share is non-negligible. Practically, this means that while individual polls can diverge, the long-run trajectory is driven more by noise and sample design than by ideological alignment of pollsters. We can also note how the MSE values are quite high, meaning that random variation (and, for some institutes, persistent house effects) remains sizable for Lega; this reflects both the party’s volatile support and methodological heterogeneity across pollsters, and implies wider uncertainty bands and a greater risk of over-interpreting short-term swings.

The smoothed trajectory in Figure 7 is the most politically informative. After the 2018 election, Lega rose rapidly, peaking around mid-2019 at levels exceeding 35% in national voting intentions. This coincides with Salvini’s central role in the first Conte government and the “Papeete crisis” (summer 2019), when an attempted early election at the height of popularity did not materialise.

Following the crisis and the formation of Conte II with the Democratic Party and Five Star Movement, support for Lega gradually declined. The pandemic years (2020-2021) saw further erosion, as Salvini’s ambivalent stances on containment and vaccination contrasted with Mario Draghi’s technocratic leadership, which attracted broader consensus. By the 2022 general election, Lega had fallen below 10%, overshadowed by Giorgia Meloni’s Fratelli d’Italia. The downward trend continued into 2023-2024 with stable single-digit levels, consistent with a reduced agenda-setting capacity within the Meloni government and internal competition in the centre-right coalition. The width of credible intervals tracks polling density: they shrink in cam-

Estimated share of Lega over time

Binomial state-space model – smoothed estimate via importance sampling

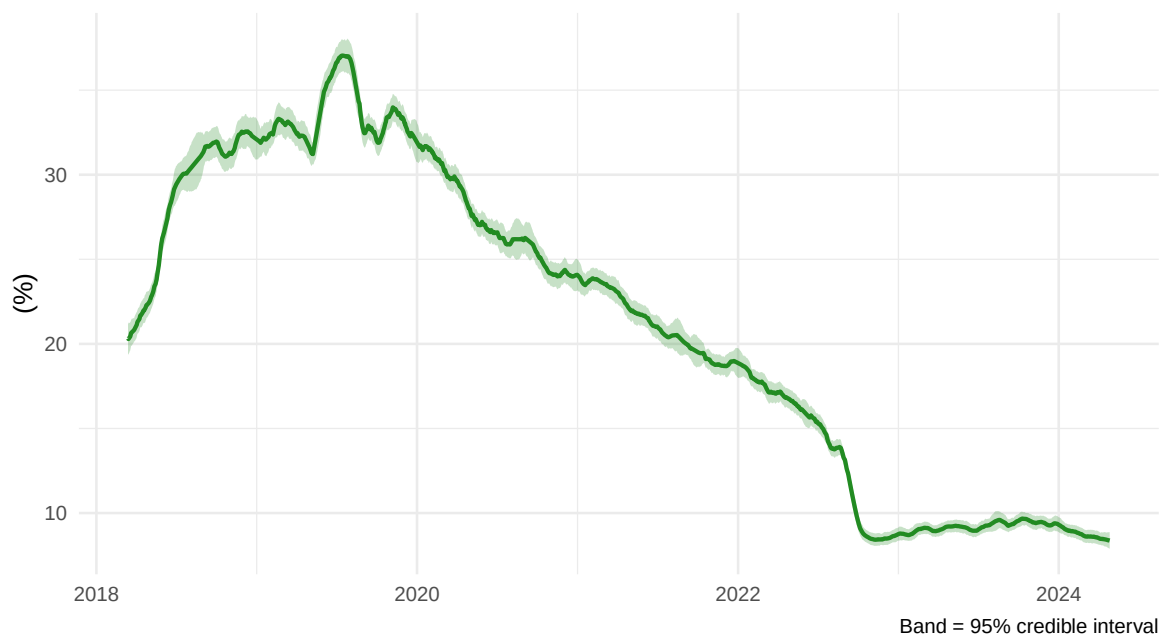


Figure 7: Estimated vote share for Lega over time (binomial model, 95% CI).

campaign periods with many surveys and widen in quieter stretches; the downward profile remains robust across specifications.

Finally, Figure 8 compares model-based house effects with realized biases at the 2022 election. The alignment is reasonably good: institutes that overestimated Lega over the long run also tended to report higher-than-actual figures immediately before the vote, and vice versa. This coherence suggests that the state-space model separates persistent methodological differences from random noise meaningfully. Notably, most pollsters overestimated Lega in the pre-election polls of 2022.

3.1.2 Partito Democratico (PD)

Partito Democratico (PD) is a major centre-left, social-democratic party in Italy, created in 2007 from a merger of Democratici di Sinistra and La Margherita. It is led by Secretary Elly Schlein and typically anchors the broader centre-left coalition. Ideologically it promotes social-democratic, progressive policies and a pro-European stance. At EU level it belongs to the Party of European Socialists, with its MEPs sitting in the S&D group. For PD, institute effects and temporal dynamics are more

Actual vs estimated bias – Lega

Last poll before the vote (Sep 25, 2022) vs house effect

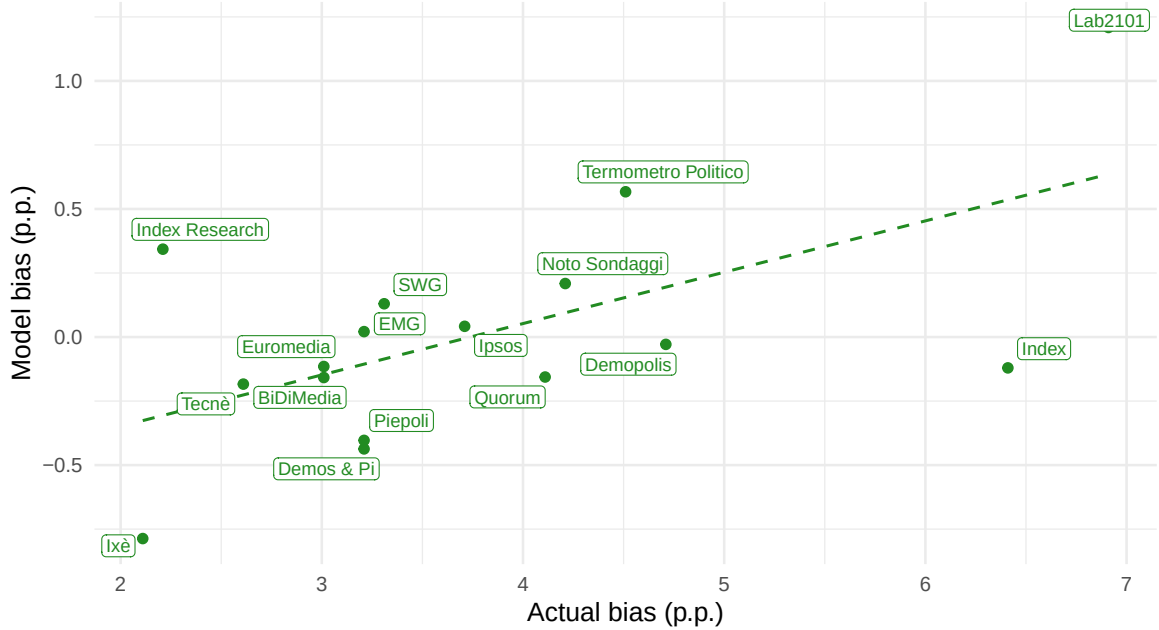


Figure 8: Model-based vs. realized bias (2022 election) for Lega.

moderate than for Lega, yet they still reveal consequential patterns.

As shown in Figure 9, most polling organisations report values close to the common trend, with deviations rarely exceeding ± 1 percentage point. Only a few institutes tend to meaningfully depart from zero: Noto Sondaggi falls below -0.5 pp, while Ixè and BiDiMedia display positive shifts. These differences are smaller than those observed for Lega, suggesting that the PD electorate is measured more consistently across institutes; nonetheless, persistent shifts of even ~ 1 pp can still shape expectations near elections.

Figure 10 indicates heterogeneity in institute stability, though at generally slightly lower levels than for Lega. Some pollsters exhibit higher estimated variances (wider dispersion around the latent trend), while others are comparatively stable. This may reflect PD's position as a mainstream party with a steadier baseline; it also means that modest changes in questionnaire design or weighting can still produce visible differences.

According to Figure 11, variance dominates the MSE for most institutes, even

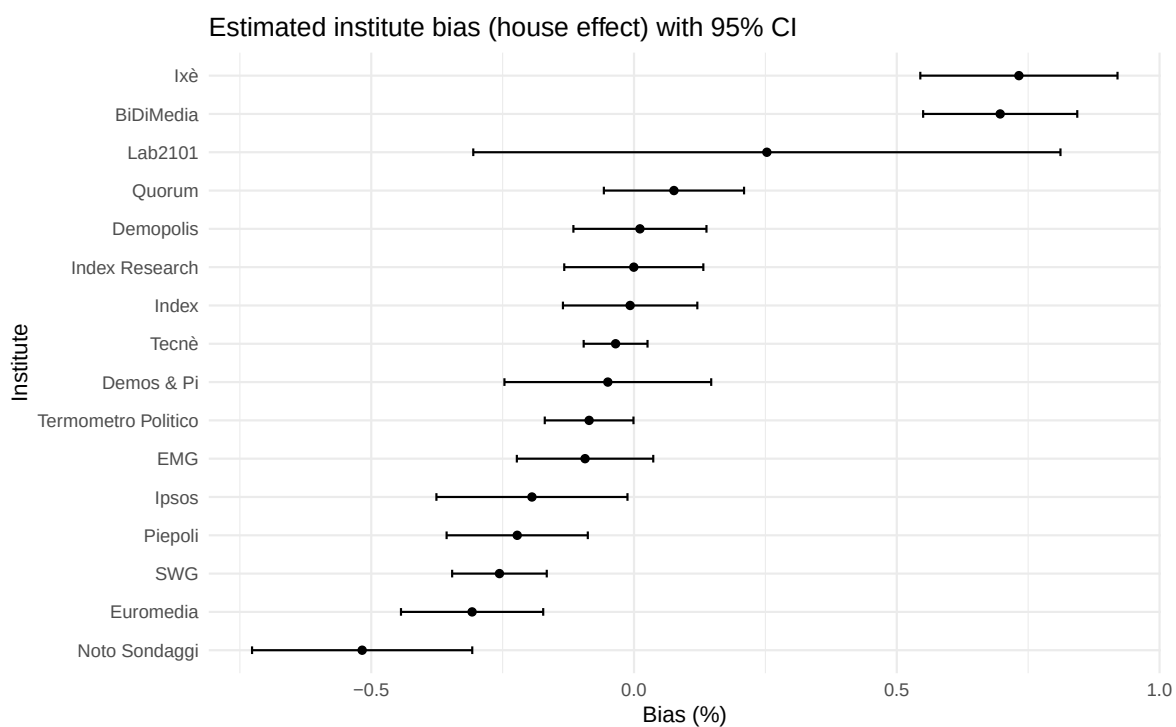


Figure 9: Estimated institute bias for the PD (point estimate and 95% CI).

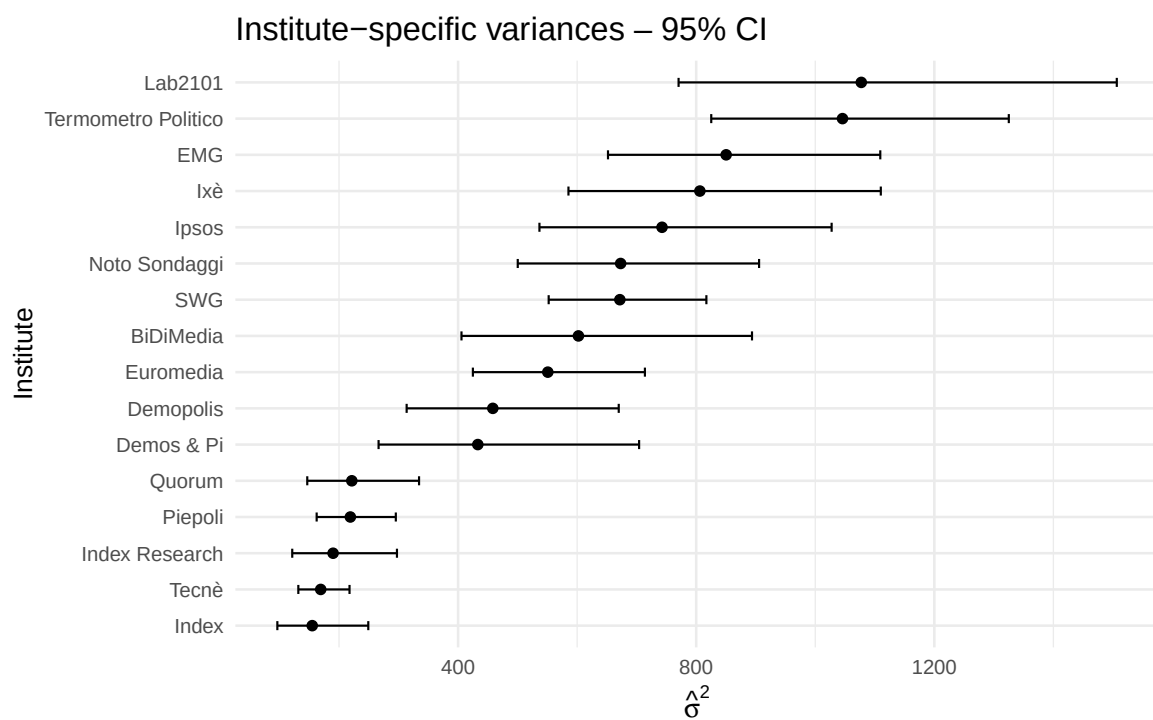


Figure 10: Institute-specific variances (95% CI) for the PD under the Gaussian model.

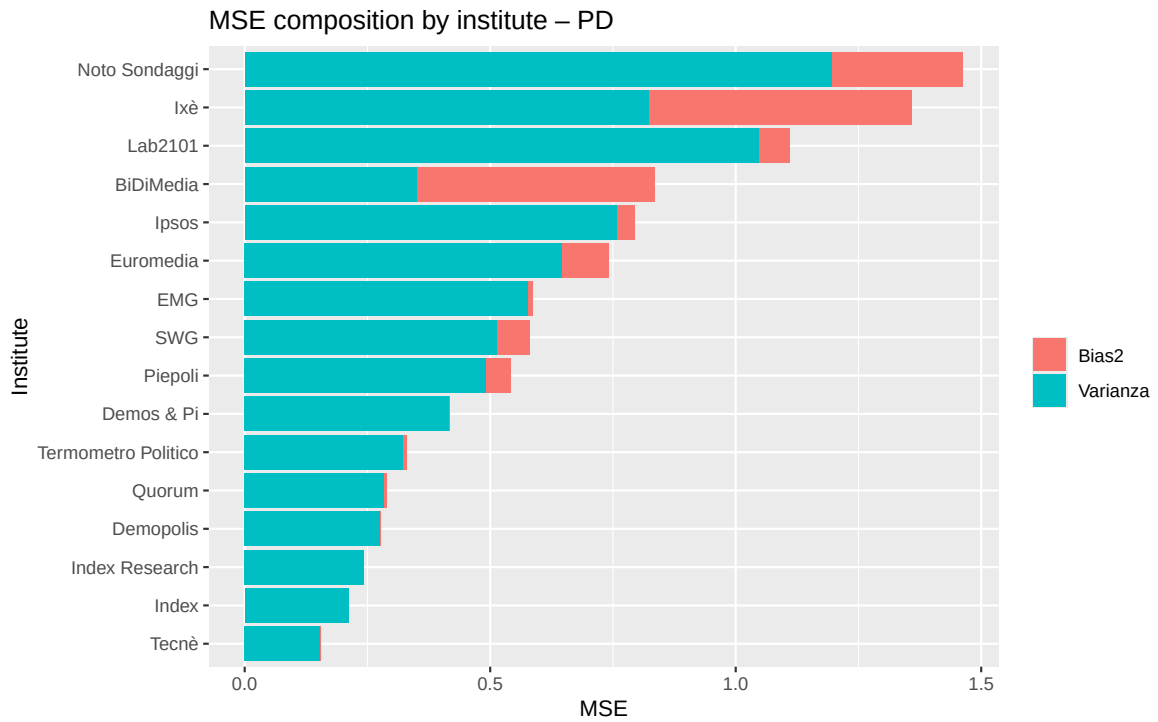


Figure 11: MSE decomposition into $bias^2$ and variance for the PD.

more than with Lega, although the absolute MSE levels for PD are lower on average. For a few pollsters, notably Ixè and BiDiMedia, the squared-bias share is more substantial, signalling a systematic positive tilt. Overall, the average poll for PD is fairly accurate, but non-negligible positive house effects arise for specific organisations.

The smoothed trajectory in Figure 12 reflects PD’s gradual repositioning. After the 2018 election the party hovered in the low 20s, stabilising during Nicola Zingaretti’s leadership and the Conte II government (2019-2021). Under the Draghi cabinet (2021-2022) PD consolidated as the main pro-European force, with estimates around 20-22%. In the run-up to the 2022 election under Enrico Letta, PD remained broadly stable while Fratelli d’Italia surged; credible intervals narrow in this period due to dense polling. Following the defeat and leadership change to Elly Schlein (spring 2023), the model indicates a modest recovery, with PD edging toward the mid-20s in 2024, consistent with a clearer opposition role and outreach to younger/progressive constituencies.

Figure 13 compares model-based house effects with realized biases in Septem-

Estimated share of PD over time

Binomial state-space model – smoothed estimate via importance sampling

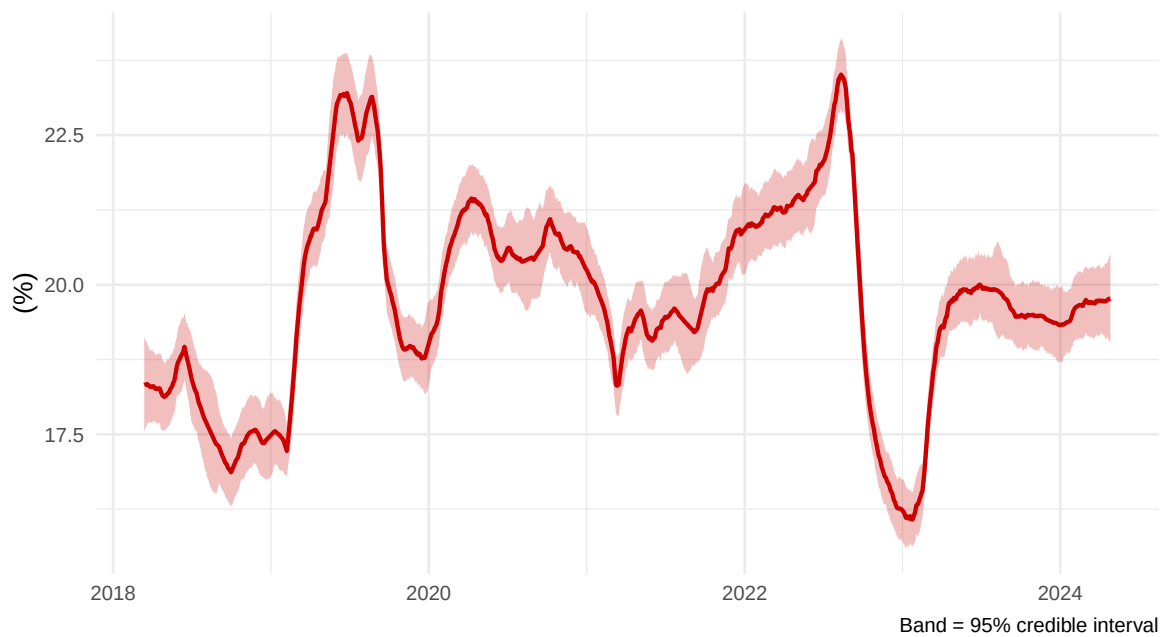


Figure 12: Estimated vote share for the PD over time (binomial model, 95% CI).

Actual vs estimated bias – PD

Last poll before the vote (Sep 25, 2022) vs house effect

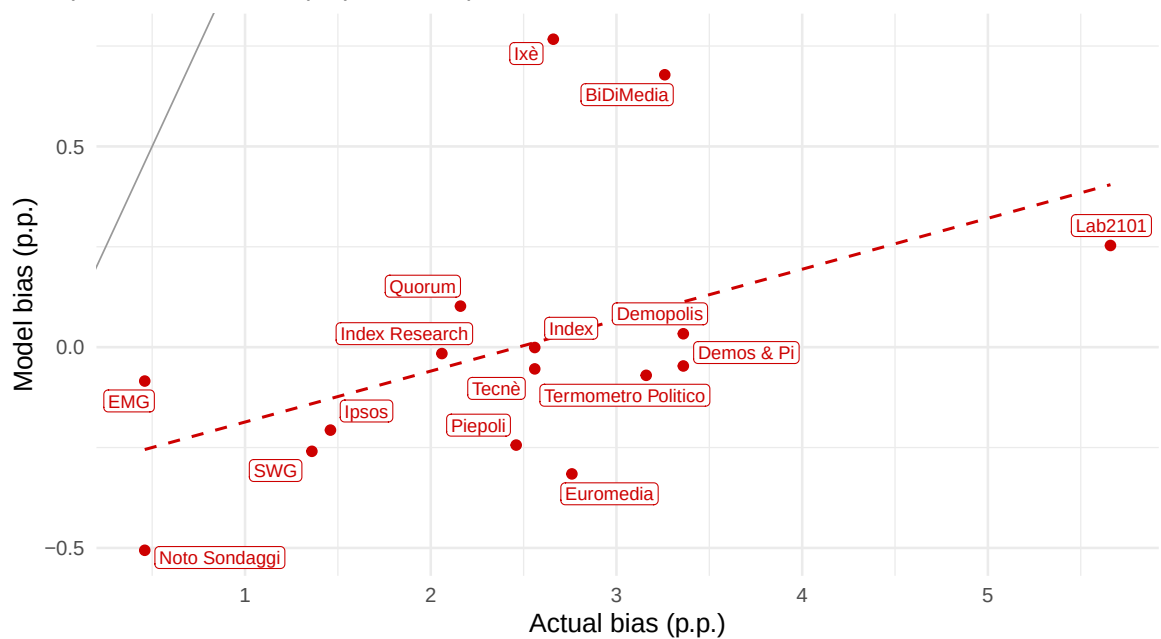


Figure 13: Model-based vs. realized bias (2022 election) for the PD.

ber 2022. The alignment is fairly strong. Ixè and BiDiMedia, which show positive modelled biases, are among those with higher pre-election overestimation; more generally, most pollsters overestimated PD before the 2022 election (actual biases are predominantly positive).

3.1.3 Fratelli d'Italia (FdI)

Fratelli d'Italia (FdI) is a right-wing to far-right, national-conservative party led by Giorgia Meloni, Italy's prime minister since 22 October 2022. Founded in December 2012 by Meloni, Ignazio La Russa, and Guido Crosetto after a split from Popolo della Libertà, it inherits the post-fascist legacy of Alleanza Nazionale. It promotes national sovereignty, social conservatism, and stricter immigration controls, and has led the centre-right government since 2022. At EU level, FdI belongs to the European Conservatives and Reformists family, with its MEPs sitting in the ECR group.

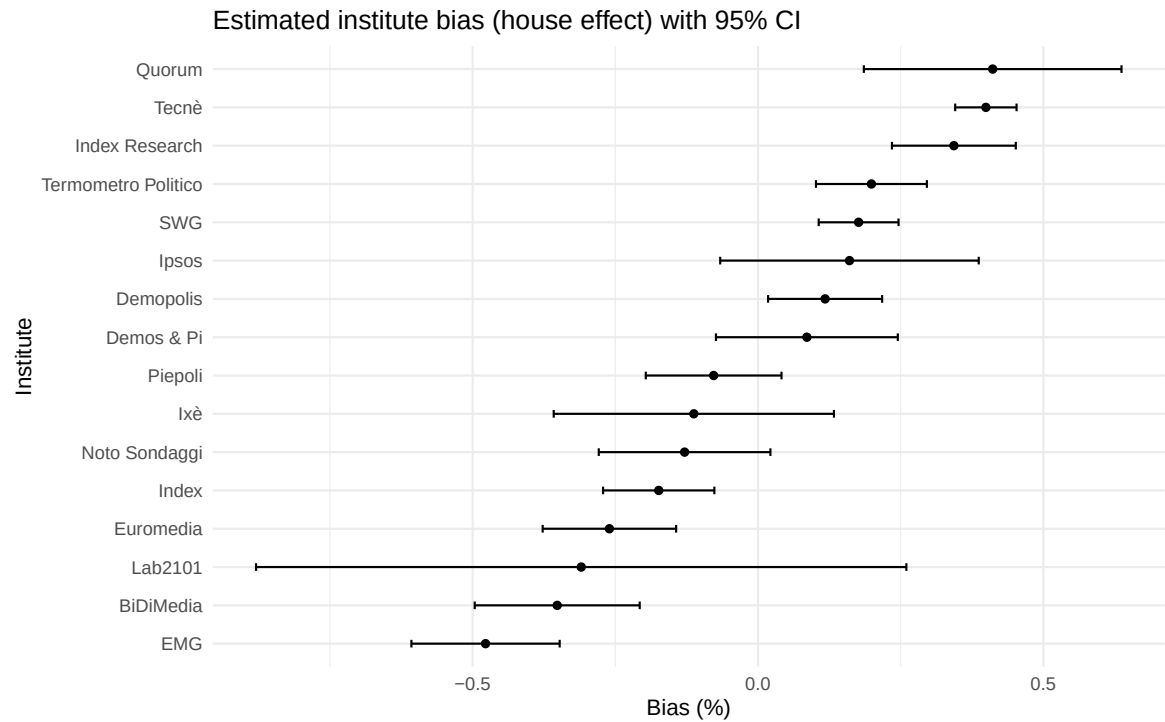


Figure 14: Estimated institute bias for FdI (point estimate and 95% CI).

Figure 14 shows modest house effects, with most institutes close to zero. Underestimation beyond -0.5 pp is limited (EMG), and only Quorum shows a positive

shift beyond +0.5 pp. The small magnitudes suggest that FdI’s remarkable growth is measured consistently across pollsters.

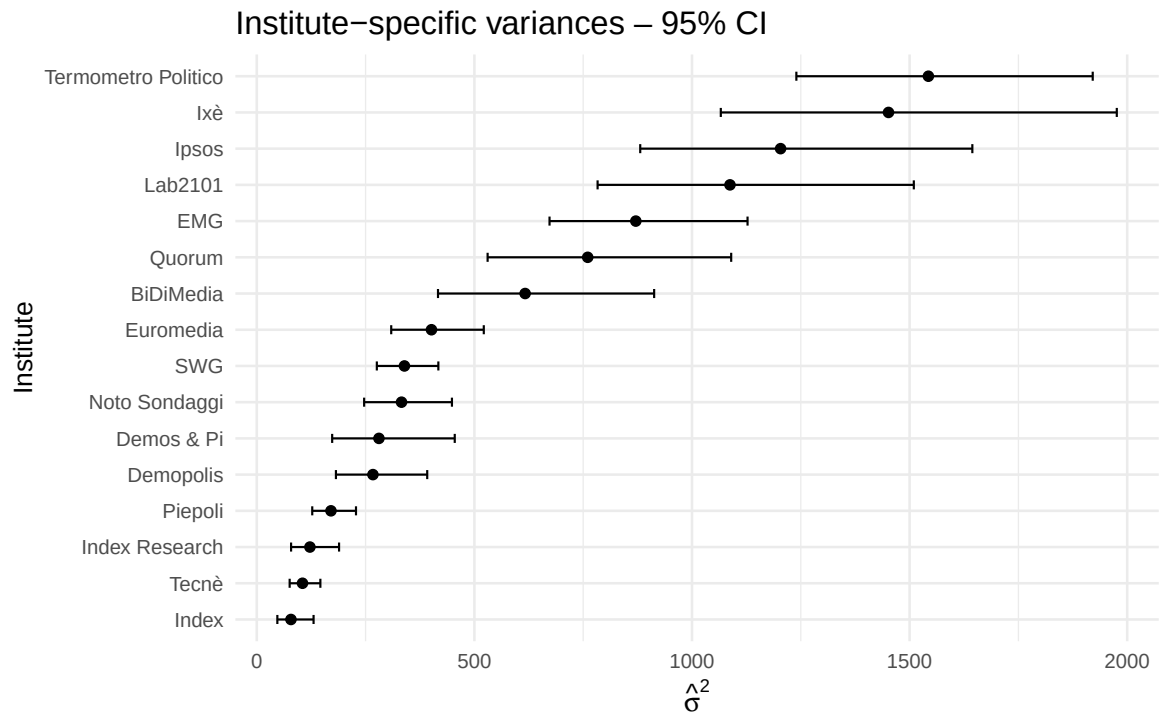


Figure 15: Institute-specific variances for FdI (Gaussian model, 95% CI).

Figure 15 highlights differences in stability: some pollsters (Tecnè, Index) exhibit lower estimated variance, whereas others (Ixè, Termometro Politico) are more fluctuant. Overall dispersion sits between that observed for Lega (higher) and PD (lower), and reflects genuine methodological diversity (sample sizes, fieldwork modes, weighting).

The MSE breakdown in Figure 16 indicates variance as the dominant component across institutes. Unlike Lega and, to a lesser extent, PD, where some bias effects are visible, FdI’s errors are driven mainly by random fluctuation. EMG displays a comparatively larger bias component (consistent with its negative shift); Tecnè shows a noticeable bias share as well, although this must be interpreted jointly with its comparatively low variance. In absolute terms, MSE levels are low (similar to PD and lower than Lega), suggesting that institutes tracked FdI’s trend well.

Figure 17 captures the extraordinary rise of FdI from residual levels (4-5%)

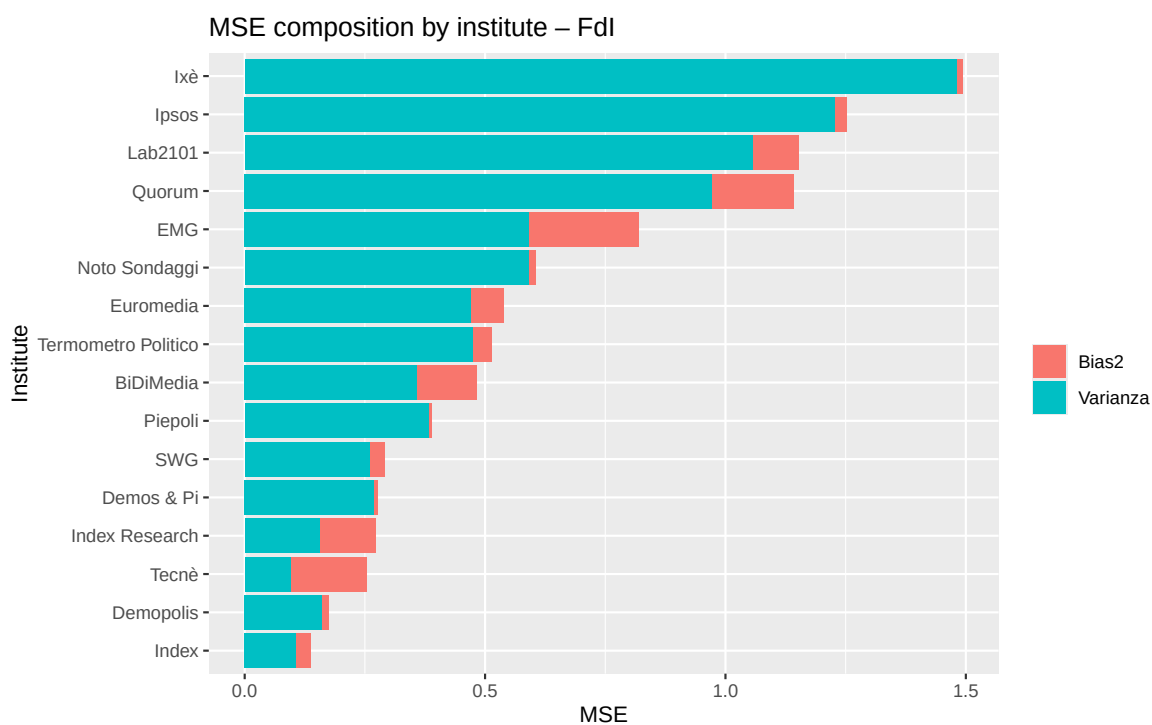


Figure 16: Decomposition of MSE into bias^2 and variance for FdI.

Estimated share of FdI over time

Binomial state-space model – smoothed estimate via importance sampling

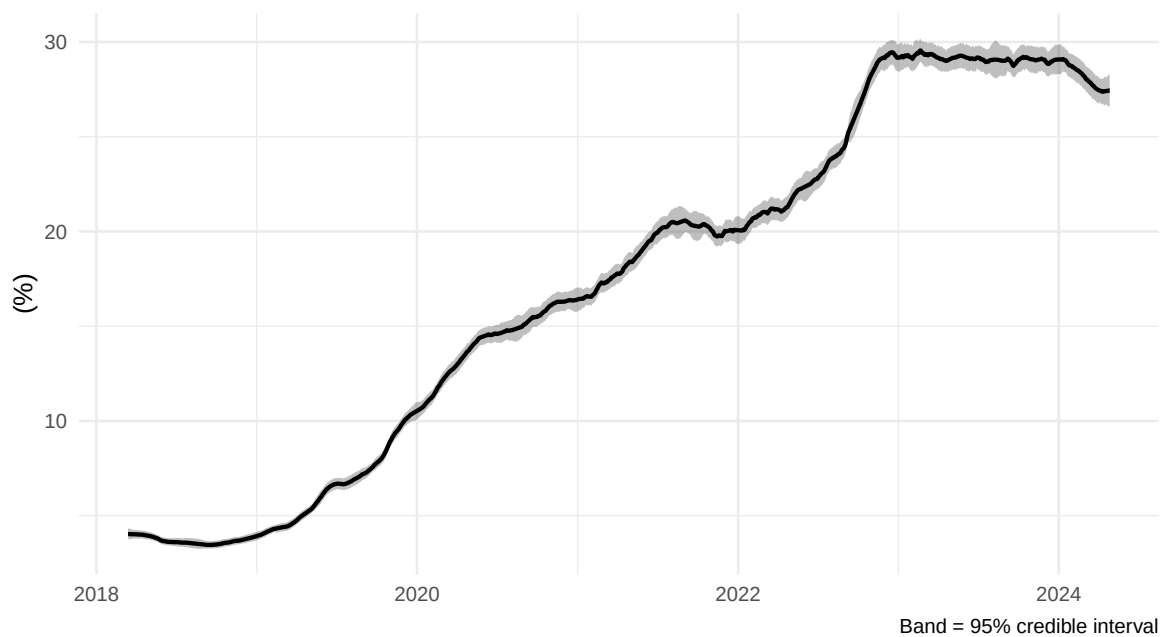


Figure 17: Estimated vote share for FdI over time (binomial model, 95% CI).

after 2018 to becoming Italy’s leading party in 2022. The increase accelerates after the first Conte government, capitalising on Lega’s decline, and consolidates during opposition to the Draghi cabinet. Narrow intervals from 2020 onward reflect dense, consistent polling. By the 2022 election, the model places FdI around 25%, close to the official outcome.

Actual vs estimated bias – FdI

Last poll before the vote (Sep 25, 2022) vs house effect

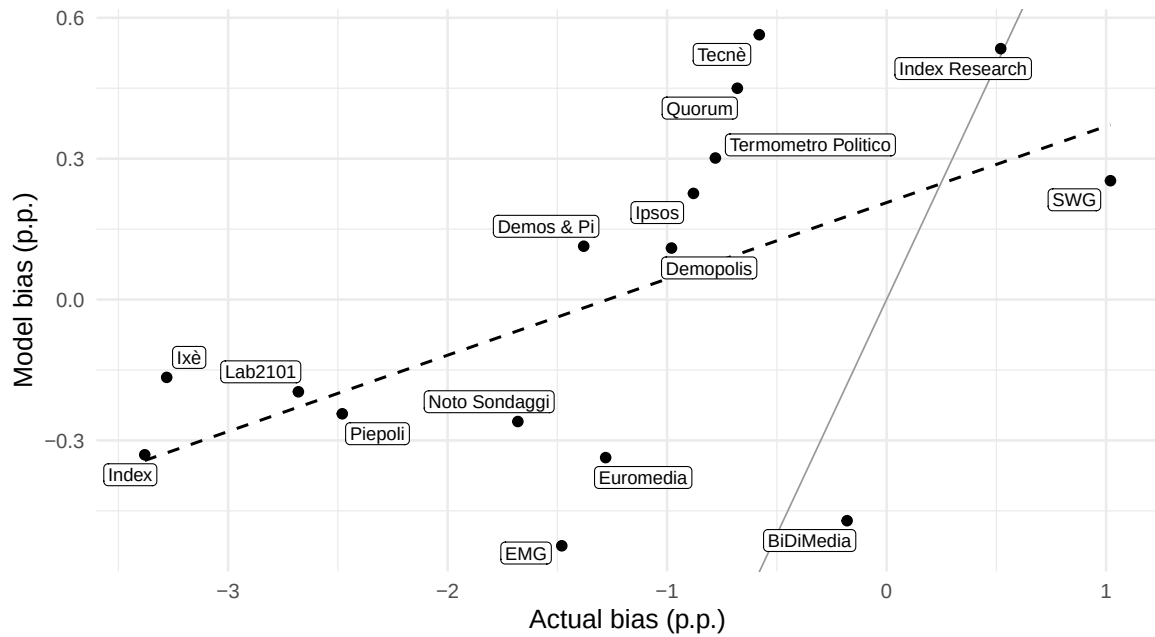


Figure 18: Model-based vs. realized bias (2022 election) for FdI.

Finally, Figure 18 compares model-based house effects with realized bias at the 2022 election. Institutes that reported below-trend values in the time series also underestimated the party in their last polls, and vice versa. Most actual biases are negative (underestimation), reinforcing the view that the late surge was strong and broad-based across pollsters.

3.1.4 Movimento 5 Stelle (M5S)

Movimento 5 Stelle (M5S) is an Italian anti-establishment, populist party founded in 2009 by Beppe Grillo and Gianroberto Casaleggio. It has been led by former prime minister Giuseppe Conte since August 2021. Ideologically it champions direct (digital)

democracy, environmentalism, and anti-corruption, evolving toward a more progressive profile. In the European Parliament, its MEPs sit with The Left group (initially as observers in July 2024, confirmed as full members on 4 February 2025).

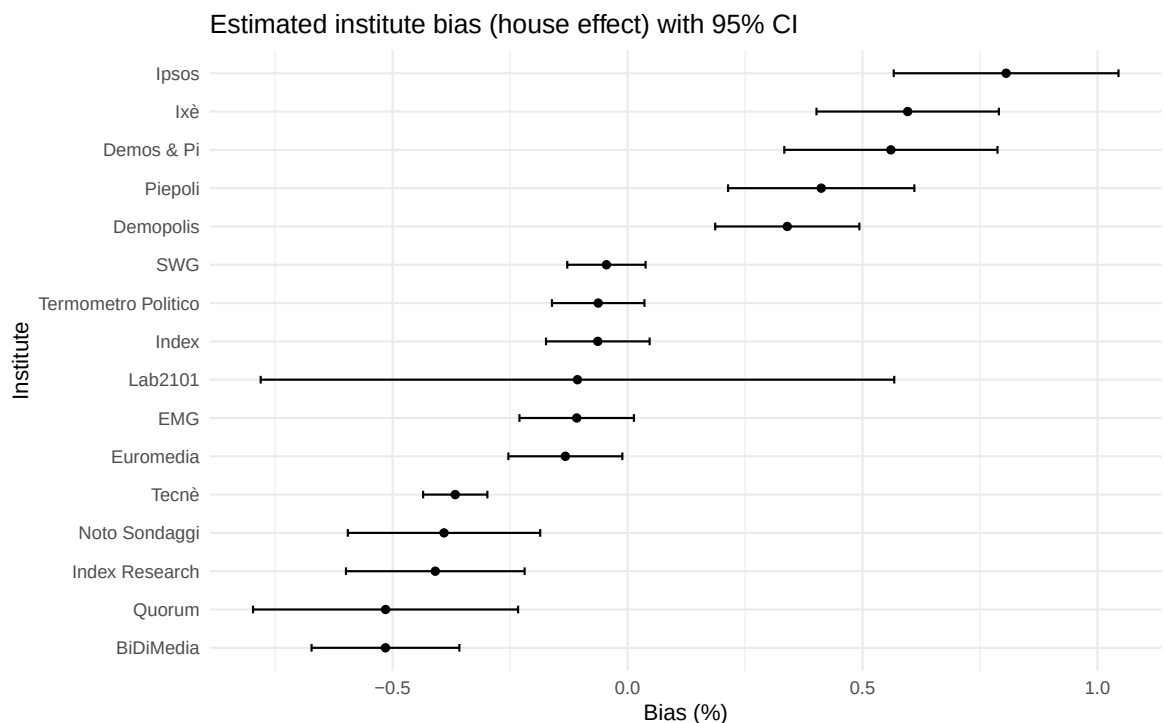


Figure 19: Estimated institute bias for M5S (point estimate and 95% CI).

Figure 19 displays a relatively balanced distribution around zero. BiDiMedia and Quorum tend to slightly underestimate M5S, whereas Ipsos, Ixè, and Demos & Pi lean slightly positive. Effects are small in size, but they become more visible against the backdrop of M5S’s marked volatility.

As shown in Figure 20, estimated variances are higher overall than for PD or FdI, closer to Lega’s levels. Institutes such as Tecnè and Index show lower variability, while others (Ipsos, Termometro Politico) appear more volatile. This instability plausibly mirrors the heterogeneous composition of the M5S electorate, which complicates precise estimation across survey designs.

Figure 21 shows variance dominating the MSE for most institutes, indicating that differences are largely driven by sampling variability rather than persistent mis-calibration. Ipsos, Ixè, Demos & Pi, and BiDiMedia exhibit a comparatively larger

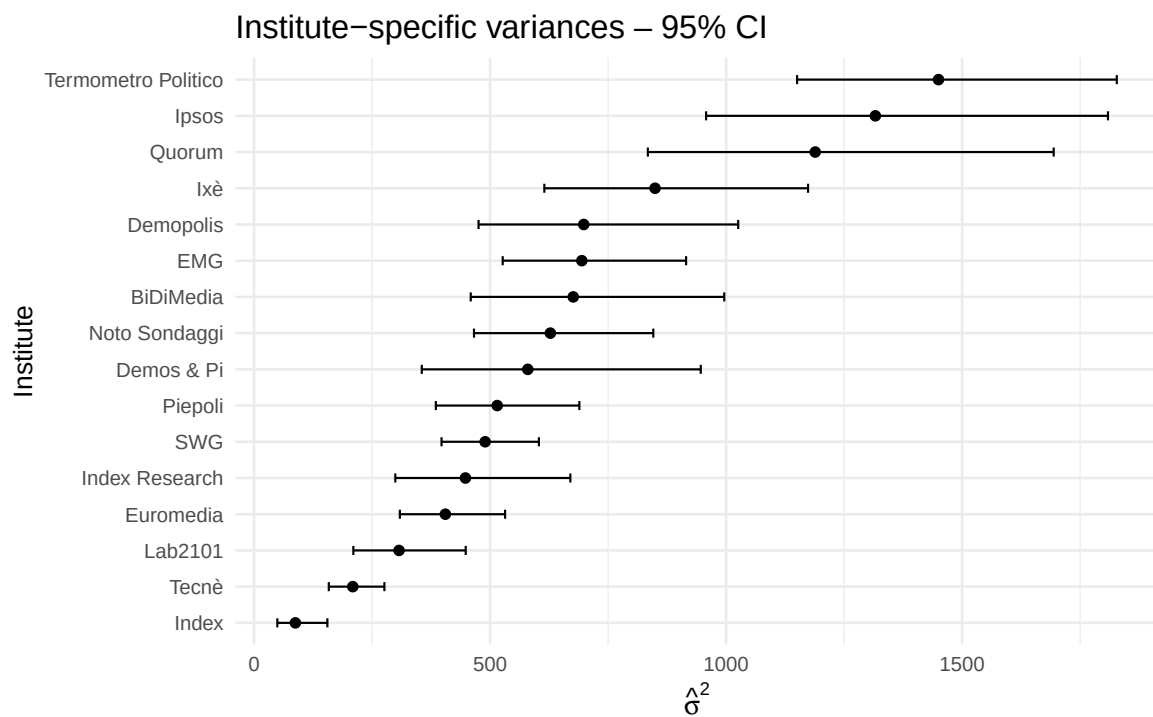


Figure 20: Institute-specific variances for M5S (Gaussian model, 95% CI).

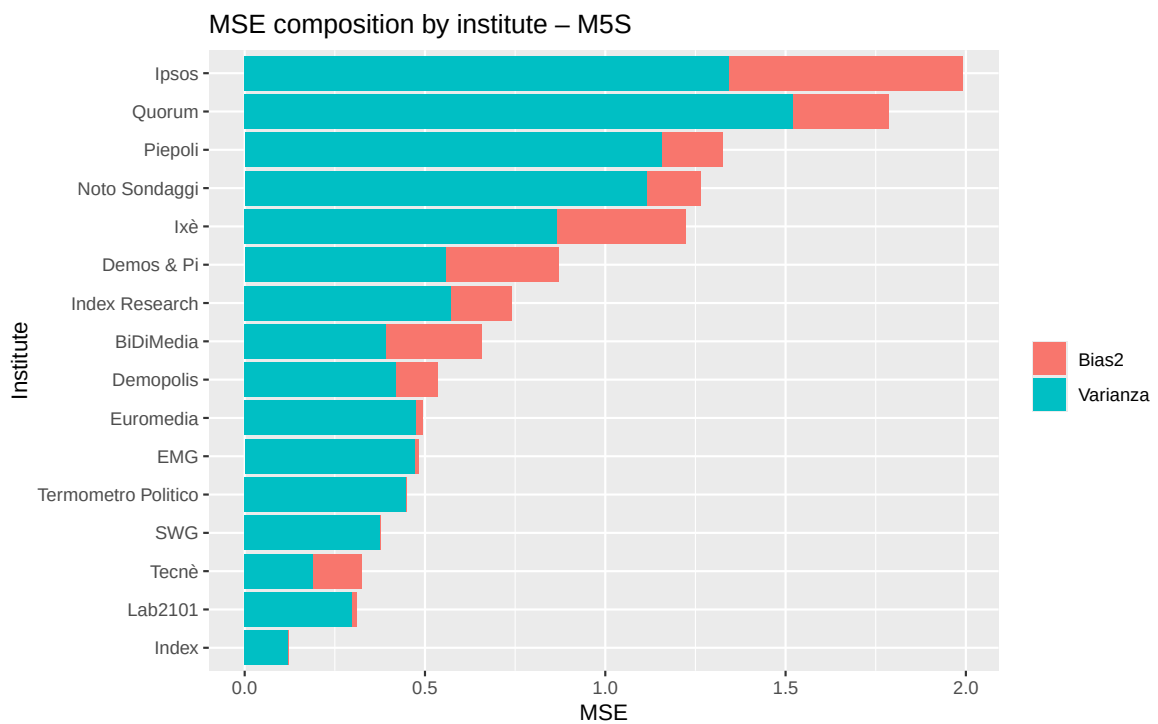


Figure 21: Decomposition of MSE into bias² and variance for M5S, by institute.

bias share. Absolute MSE levels are relatively high (close to Lega's), consistent with the more turbulent trend for both parties over these years.

Estimated share of M5S over time

Binomial state-space model – smoothed estimate via importance sampling

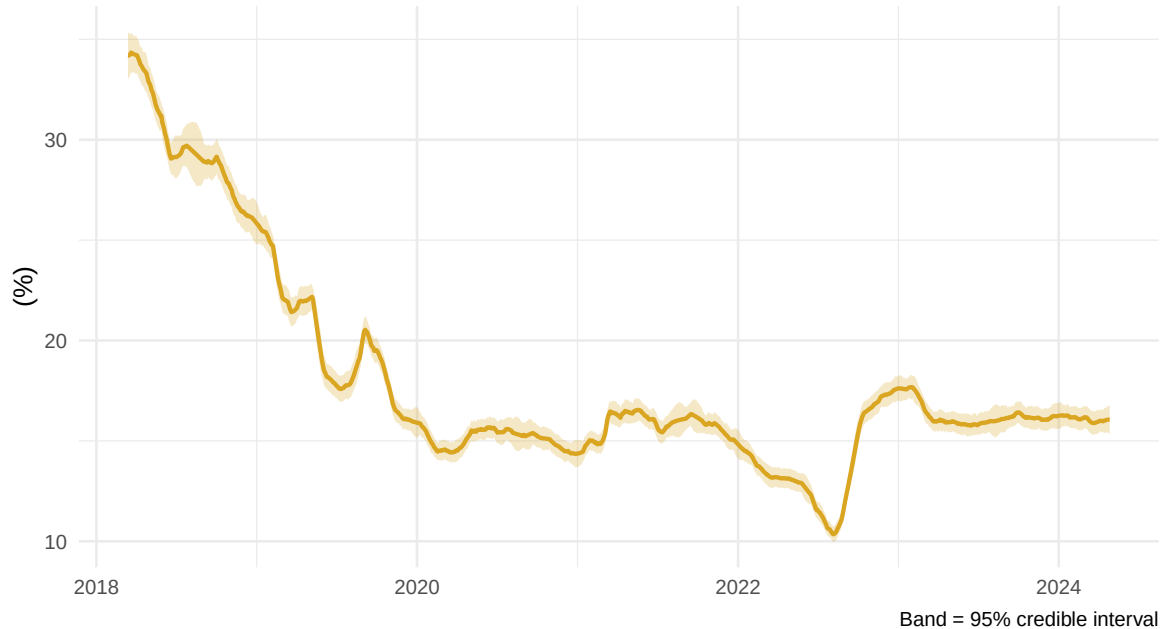


Figure 22: Estimated vote share for M5S over time (binomial model, 95% CI).

The trajectory in Figure 22 illustrates the sharp decline from nearly 33% after 2018, accelerating after the Conte I split with Lega and continuing under Conte II. The Draghi period (2021-2022) saw further contraction amid internal tensions. In the run-up to 2022, M5S rebounded to roughly 15%, attributed in part to Conte's personal campaign (focused on the defence of “reddito di cittadinanza”). Post-2022, the party stabilises around 15-17% with relatively wide but steady intervals, indicating consolidation of a smaller, more loyal base.

Finally, Figure 23 compares model-based house effects with realized bias at the 2022 election. Alignment is weaker than for FdI or PD: several institutes underestimated M5S immediately before the vote, while others slightly overestimated it. This pattern is consistent with the difficulty of capturing a late recovery during the campaign.

Actual vs estimated bias – M5S

Last poll before the vote (Sep 25, 2022) vs house effect

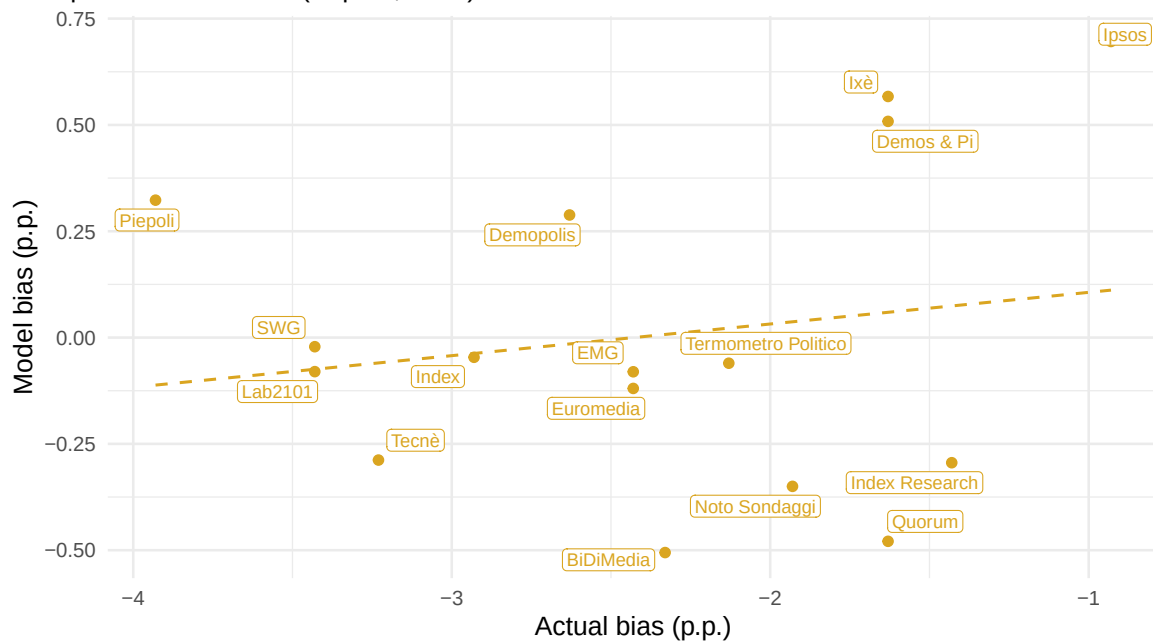


Figure 23: Model-based vs. realized bias (2022 election) for M5S.

3.1.5 Forza Italia (FI)

Forza Italia (FI) is a centre-right, liberal-conservative and Christian-democratic party founded by Silvio Berlusconi in 1994 and re-established in 2013. It is led by Antonio Tajani (elected national secretary on 15 July 2023), who also serves as Deputy Prime Minister and Foreign Minister. FI is a partner in Italy's centre-right coalition government led by Giorgia Meloni since October 2022. Ideologically, it is pro-European, pro-business, and strongly Atlanticist. In the European Parliament, its MEPs belong to the European People's Party (EPP) and sit with the EPP Group.

Figure 24 shows somewhat wider dispersion than for larger parties. A few institutes (Lab2101, Index Research, SWG) tend to report lower values, while others (like Tecnè, Ixè, Piepoli) often overestimate the party. Given FI's modest average support, deviations occasionally above ± 1 pp can materially affect perceived momentum.

As shown in Figure 25, institutes differ considerably in stability. Because FI's electorate is relatively small, even modest design differences (sample size, quotas, panel composition) can translate into visible percentage-point fluctuations.

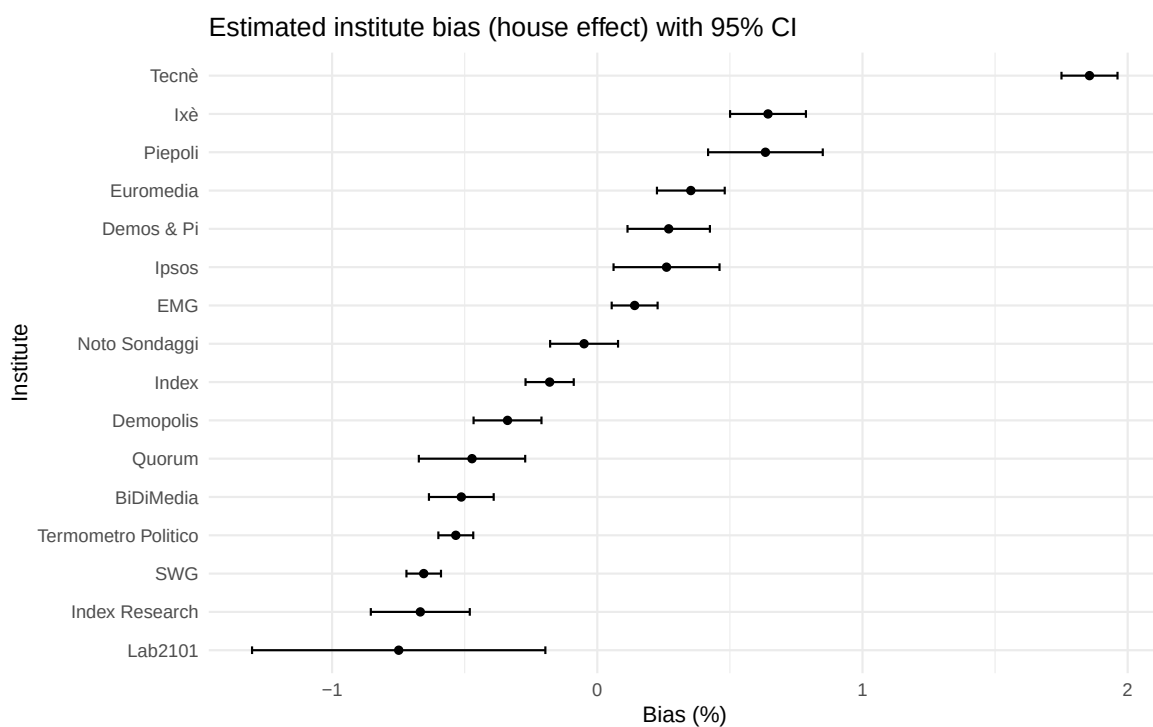


Figure 24: Estimated institute bias for FI (point estimate and 95% CI).

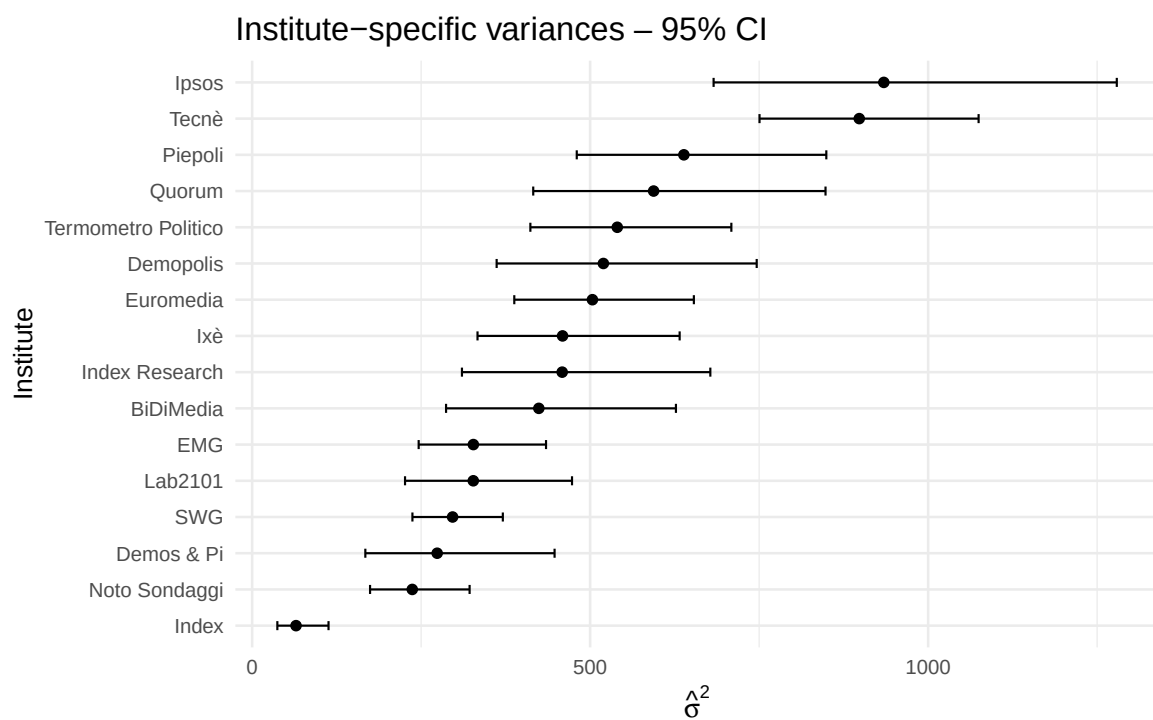


Figure 25: Institute-specific variances for FI (Gaussian model, 95% CI).

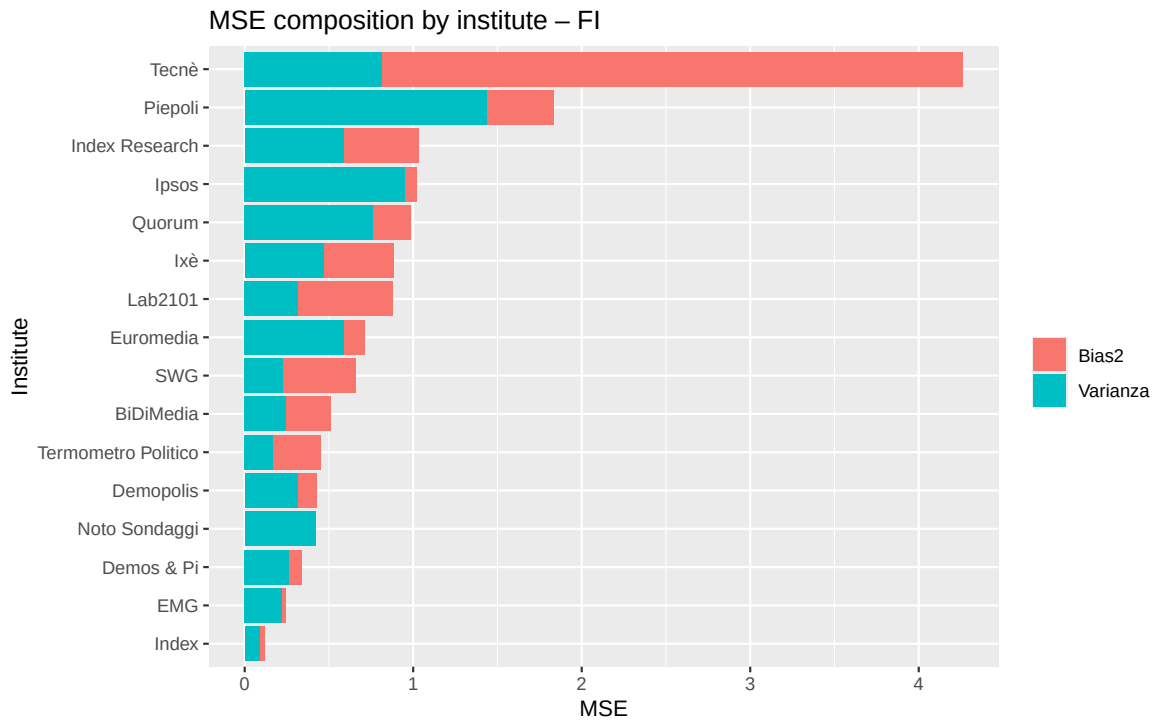


Figure 26: MSE decomposition into bias² and variance for FI, by institute.

Figure 26 indicates that for FI, both variance and bias contribute meaningfully to MSE, more so than for FdI or PD (and partly M5S). Some institutes (like Piepoli) show relatively high MSE with a notable bias component, while others are closer to the average. One institute, Tecnè, exhibits a particularly large MSE driven mainly by a positive bias (overestimation). Given the potential for selection and mode effects related to audiences and commissioning outlets, these results should be interpreted with appropriate caution and with attention to transparency in methods.

Figure 27 shows FI's gradual decline from roughly 13% in 2018 to about 8-9% by the 2022 election. Generational turnover and the waning centrality of Berlusconi's leadership, alongside competition within the centre-right (FdI and Lega), likely contributed to this contraction. After Berlusconi's death in 2023, the model suggests continued stagnation: any short-lived sympathy bump fades, with FI remaining below 10% and largely dependent on coalition dynamics. Finally, Figure 28 shows a relatively strong correspondence between model-based and realized bias in 2022: notably, Tecnè registers a pronounced overestimation (the only sizeable positive realized bias alongside Piepoli), consistent with the modelled house effect.

Estimated share of FI over time

Binomial state-space model – smoothed estimate via importance sampling

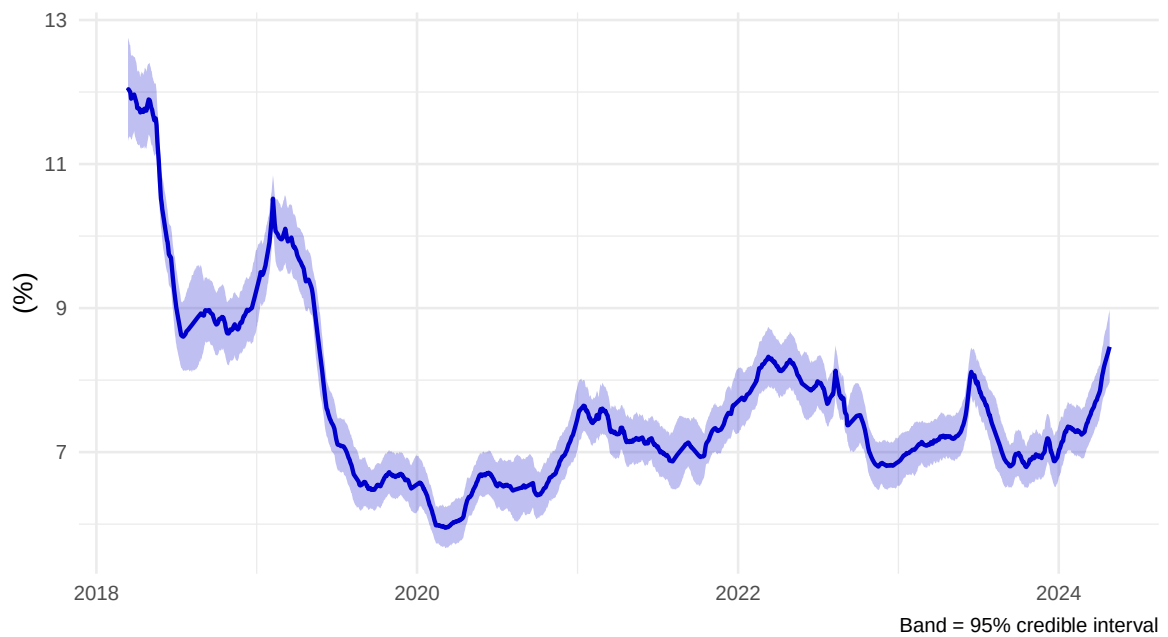


Figure 27: Estimated vote share for FI over time (binomial model, 95% CI).

Actual vs estimated bias – FI

Last poll before the vote (Sep 25, 2022) vs house effect

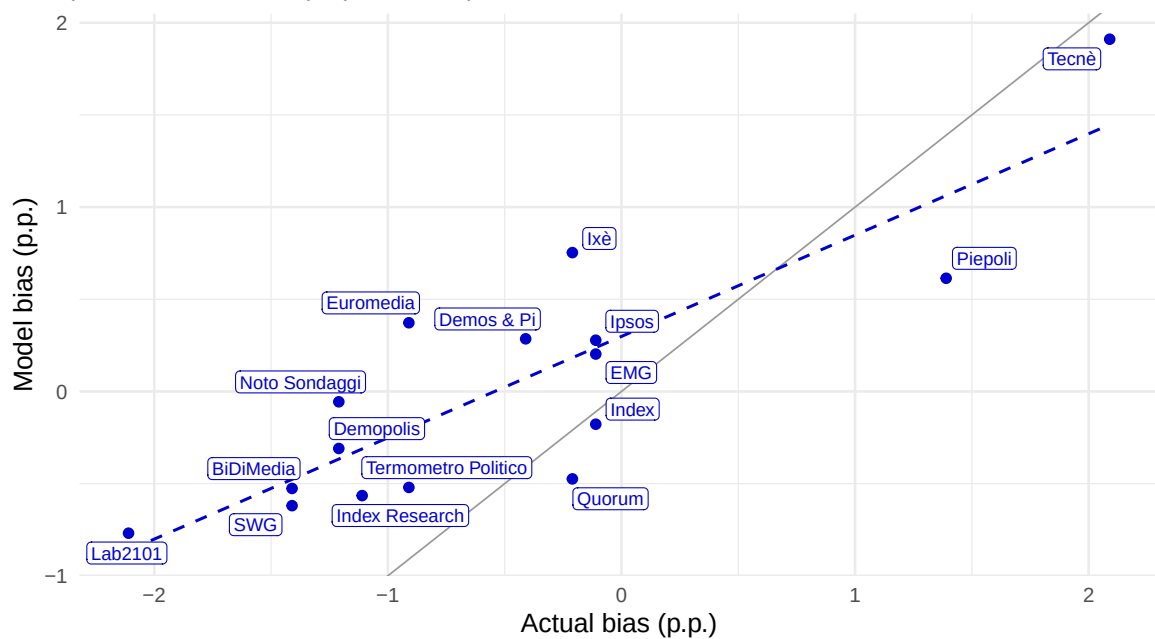


Figure 28: Model-based vs. realized bias (2022 election) for FI.

3.2 Institute-level Analysis

This subsection synthesises the evidence *by institute*, aggregating results across the five parties.

3.2.1 Cross-institute and cross-party comparison

This section synthesises the cross-institute evidence accumulated so far. We proceed from a compact, all-in-one heatmap of biases, to party-by-party bias panels, and finally to the MSE decomposition into *bias*² and variance (as defined in Section 2.3.1).

Figure 29 provides an overall view of the sign and significance of institute biases across the five major parties. Several robust patterns align with the party chapters in Section 3:

- Most cells are close to zero (white), but non-negligible tilts appear for some institutes. For example, Lab2101 and Termometro Politico tend to overestimate Lega (Section 3.1), while Ixè, Demos & Pi, and Piepoli underestimate it. For PD, Ixè and BiDiMedia lean positive whereas Noto Sondaggi is slightly negative (Section 3.2). FdI shows only mild effects, with Quorum slightly positive and EMG negative (Section 3.3). For M5S, Ipsos and Ixè are positive while BiDiMedia and Quorum are negative (Section 3.4). FI is the most dispersed, with Tecnè and Piepoli on the positive side and Lab2101, Index, and SWG negative (Section 3.5).
- Smaller parties, especially FI, display more frequent significant cells. A shift of 0.5-1 pp represents a larger fraction of their average support, so differences are more consequential in relative terms.
- Very few institutes are consistently positive or negative across all parties. This supports the view that house effects are largely party-specific, rooted in sampling composition or weighting interacting with the electorates rather than in uniform institute-wide miscalibration.

Bias per institute and party

Red: under-estimate • Green: over-estimate • White: IC95% doesn't include 0



Figure 29: Heatmap of estimated house effects by institute (rows) and party (columns). Red = under-estimation, green = over-estimation, white = not statistically significant at 95%.

In sum, no party is systematically over- or under-estimated by a clear majority of institutes. Deviations are small, often below ± 1 pp, and tend to offset across organisations.

Estimated institute bias (house effect)

Points = estimate; bars = 95% CI; vertical line = 0

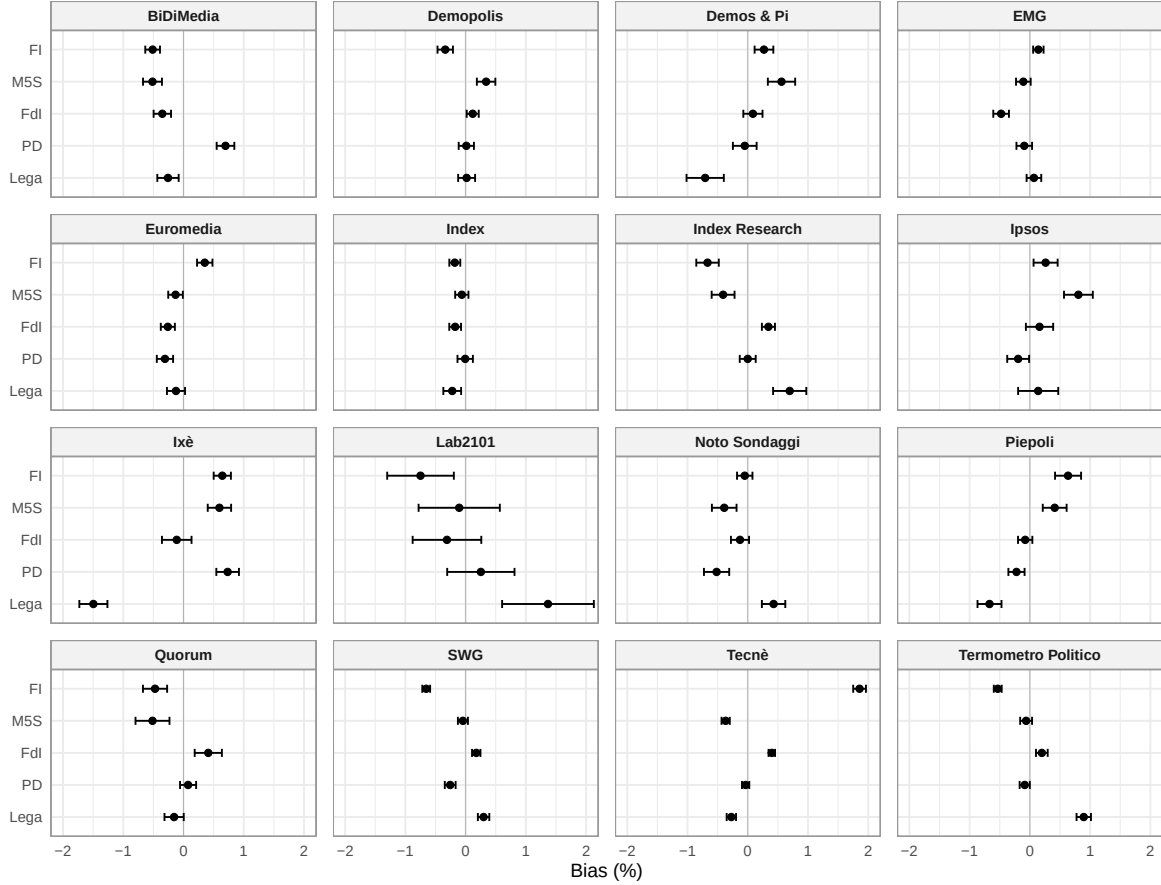


Figure 30: Estimated institute bias (points) with 95% CIs (bars), by party. The vertical line at 0 marks no bias. Estimates are reported on the percentage-point scale; binomial logit-scale effects are mapped as in Section 2.3.3.

Figure 30 quantifies magnitudes with 95% intervals. Three remarks connect these panels directly to the results chapters:

- For most institute-party pairs, intervals lie within ± 1 -2 pp. This is consistent with the detailed party analyses, where we found that differences rarely exceeded a couple of points but were significant in sign (for example Lab2101 and Termometro Politico positive for Lega, BiDiMedia positive for PD, Quorum negative for FdI).
- Some institutes show unusually wide uncertainty. Lab2101 in particular has

confidence intervals substantially larger than peers, signalling less precise bias estimation. This reflects both noisier data and relatively few available polls compared with major institutes such as Ipsos or SWG, and cautions against over-interpreting its point estimates.

- Signs align with the validation against the 2022 election. Institutes that systematically overestimated Lega or PD in the model also overestimated them in pre-election polls (Figures 8, 13). Similarly, underestimation of FdI by EMG and overestimation of FI by Tecnè and Piepoli appear consistently in both modelled biases and realised election errors.

MSE composition per institute

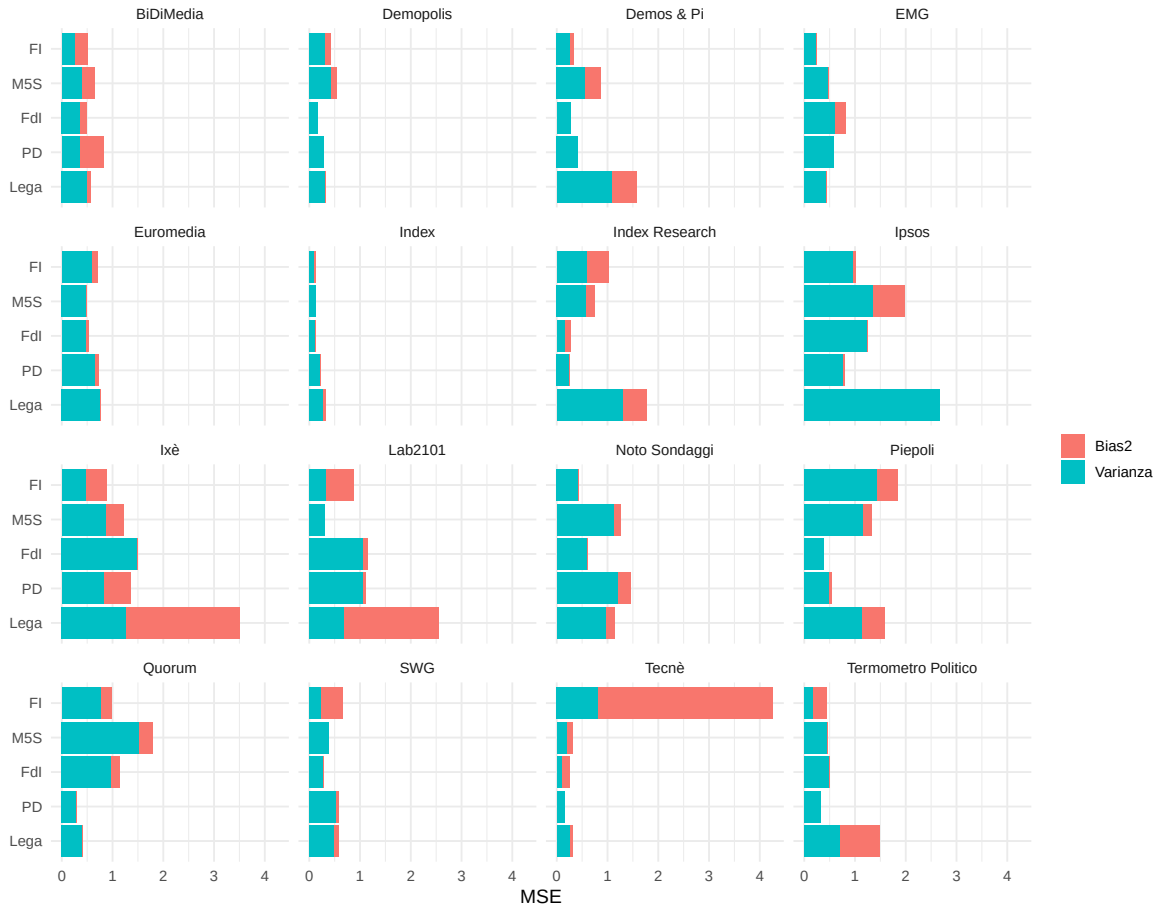


Figure 31: MSE composition by institute and party: decomposition into $bias^2$ and (average) variance component (see Section 2.3.1).

Figure 31 implements the heuristic decomposition introduced in Section 2.3.1:

$$MSE_j \approx \delta_j^2 + \mathbb{E}_t \left(\frac{\sigma_j^2}{n_{j,t}} \right),$$

which we use as an interpretive device rather than an exact identity. Looking across institutes rather than parties, Figure 31 suggests a useful typology based on the balance of variance and bias components:

- **Low MSE, variance-driven.** A first group of institutes (Demopolis, EMG, Euromedia, Index, SWG for all but FI) typically shows low mean squared errors where the variance bar dominates. Their estimates fluctuate around the latent trend but without strong directional shifts. This profile is consistent with relatively standardised survey designs, large or regular samples, and broadly neutral house effects. Differences between these institutes tend to cancel out quickly in poll averages.
- **Moderate MSE, variance-driven.** A second group (Noto, Quorum and Ipsos when excluding M5S) displays somewhat larger MSE values, still largely explained by the variance component. These institutes capture the main trajectories but with more noise, often reflecting smaller or less regular samples. Their house effects are minor but their volatility means individual polls can diverge from the consensus.
- **Low-Moderate MSE with visible bias.** A third group includes Demos & Pi, Index Research, Termometro Politico, BiDiMedia, and Piepoli. Here the MSE bars are higher and the $bias^2$ component contributes non-negligibly alongside variance. This indicates persistent, party-specific tilts (Termometro Politico positive for Lega, BiDiMedia positive for PD, Piepoli overestimating FI) which add a structural layer to the random noise. For these institutes, disagreement with the average is not purely accidental but partly systematic.
- **High MSE with strong bias.** Some institutes display particularly large MSEs

where both the variance and the $bias^2$ components contribute substantially. Ixè belongs here: although its point estimates are not extreme, the model attributes a persistent directional shift (positive on PD, negative on Lega) that adds structural error. Lab2101 also falls in this group: beyond its positive tilt for Lega, its estimates are characterised by unusually wide confidence intervals in Figure 30, indicating unstable results and magnifying both random variability and systematic bias. The large uncertainty attached to Lab2101 suggests that its polls should be weighted very cautiously in aggregate analyses.

- **Special case: Tecnè.** Tecnè is unique in that it would otherwise fall in the first group (low MSE and minimal bias) for most parties, but exhibits an exceptionally large bias for Forza Italia (Section 3.5). The tall $bias^2$ bar for FI shows that this deviation cannot be explained by sampling noise alone. One plausible explanation is linked to the commissioning environment: Tecnè’s results are frequently broadcast on Mediaset channels, owned by the Berlusconi family, which provides incentives for a systematic upward tilt in FI support. This pattern illustrates how institutional and media affiliations may shape persistent house effects in ways not reducible to survey design alone.

This classification highlights that not all institutes contribute the same type of uncertainty. For some (for example Demopolis, SWG, Ipsos) errors are mostly random and will average out in poll-of-polls. For others (for example Termometro Politico, BiDiMedia, Piepoli), small but persistent biases add systematic divergence. And in a few cases (Tecnè, Lab2101, Ixè), structural shifts dominate, so their inclusion can meaningfully tilt poll averages unless explicitly corrected. This institutional heterogeneity justifies both the state-space modelling strategy of Section 2.3 and the selective bias adjustments applied in Section 3.

It is worth stressing that these classifications are defined relative to the identifying restriction of the model, which imposes the sum of institute biases to be zero. Estimated house effects should therefore be interpreted as relative tilts with respect to the cross-institute average, not as absolute deviations from the electoral truth. Nonetheless,

the relative comparisons remain substantively meaningful: institutes with consistently wider or more directional deviations (for example Ixè, Lab2101, Tecnè for FI) exert stronger leverage on poll averages and require explicit correction or down-weighting to avoid distortions. Importantly, the party-by-party analysis in Section 3 showed a clear correlation between modelled biases and realised biases in the 2022 election, reinforcing that these relative estimates capture persistent, substantive house effects rather than mere statistical artefacts.

Overall, the figure highlights that while variance dominates the error structure for most institutes, certain organisations (notably Tecnè and Lab2101) contribute persistent structural bias on top of variance. This explains why disagreements between institutes are sometimes ephemeral noise, and sometimes systematic divergence.

3.2.2 Aggregate institute rankings and standardized comparisons

The party-by-party evidence above can be summarised in cross-party tables that rank pollsters on two diagnostics: the *mean absolute bias* (accuracy in level, irrespective of sign) and the *mean squared error* (overall error combining bias and variance). I report each metric (i) in raw, percentage-point units and (ii) after standardising *within party* before aggregation, so that parties with different scales/volatility contribute on a comparable basis.

As shown in Sections 3.1-3.5, parties differ in both trend volatility and typical poll noise. Without standardisation, parties with inherently larger dispersion can dominate the average. I therefore compute *z-scores within each party* (subtract party-specific mean and divide by party-specific standard deviation) and only then average across parties. This transforms the metric units into “within-party standard deviations”, promoting comparability across parties. (The raw charts and tables keep percentage-point interpretability.)

Graph A: Mean Absolute Bias (no transformation)

Construction. For each institute, I take the absolute value of the estimated house effect for each party (percentage points) and average across parties (unweighted).

Ranking istituti – Average Absolute Bias

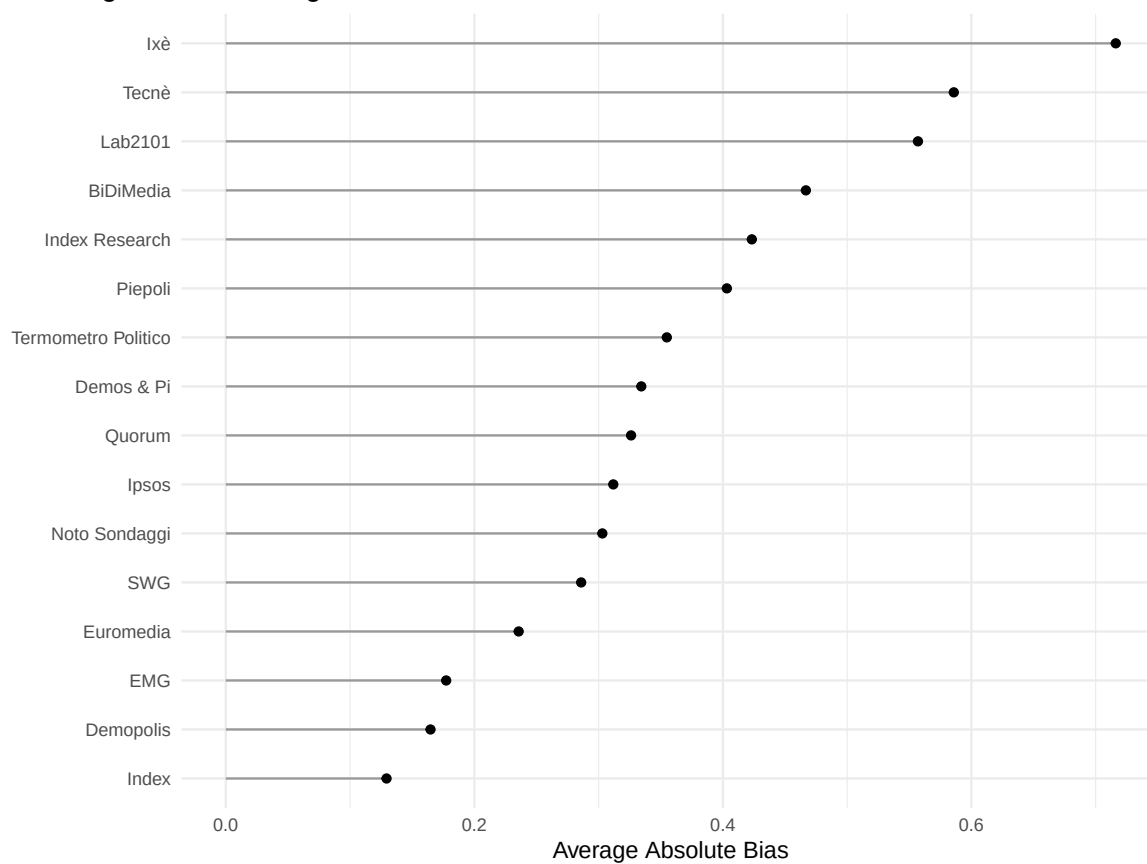


Figure 32: Aggregate ranking by mean absolute bias across parties (raw, percentage points).

What it shows. This ranking captures the typical magnitude of systematic deviation from the cross-institute trend, regardless of direction. In other words, it reflects how far each organisation tends to drift from the latent trajectory after controlling for sampling noise, with positive and negative biases treated symmetrically. Consistent with the party chapters, institutes with small absolute deviations cluster at the low end of the ranking, confirming that their house effects are minor and largely absorbed by variance (as already observed for Demopolis, Euromedia and SWG in Figures 30 and 31). By contrast, institutes repeatedly flagged in the party panels (such as Ixè with a positive tilt for PD and a negative one for Lega, or Lab2101 with wide and unstable estimates) display larger shifts here as well. Tecnè also emerges towards the upper half of the ranking: although neutral for most parties, the systematic overestimation of FI (Section 3.5 and Figure 29) raises its aggregate score, highlighting that even a single strong tilt can alter cross-party averages. This coherence reinforces the interpretation from the heatmap and MSE decomposition that house effects are mostly party-specific.

Graph B: Mean Squared Error (no transformation)

Construction. For each institute and party I compute $MSE = \text{Bias}^2 + \text{Variance}$ using the decomposition in Section 2.3.1, and then average across parties (percentage-point scale).

What it shows. Unlike the decomposition plots of Section 2.3, this graph summarises only the total MSE by institute. The pattern is consistent with earlier evidence: most institutes sit at relatively low levels (Demopolis, SWG, EMG), indicating that their overall divergence from the latent trend is limited and largely comparable. In the upper tier appear Ixè and Lab2101, whose instability and systematic tilts identified in Figures 29 and 30 translate here into higher aggregate errors. Tecnè occupies a mid-upper position: close to the neutral group for most parties, but pushed upwards by the persistent positive bias on FI (Section 3.5). Ipsos, by contrast, records a quite high MSE: although one of the most regular institutes, its estimates for M5S diverge somewhat from the cross-party average, placing it above the purely variance-driven group and really close to the most problematic pollsters. Other institutes such as Eu-

Ranking istituti – Average MSE

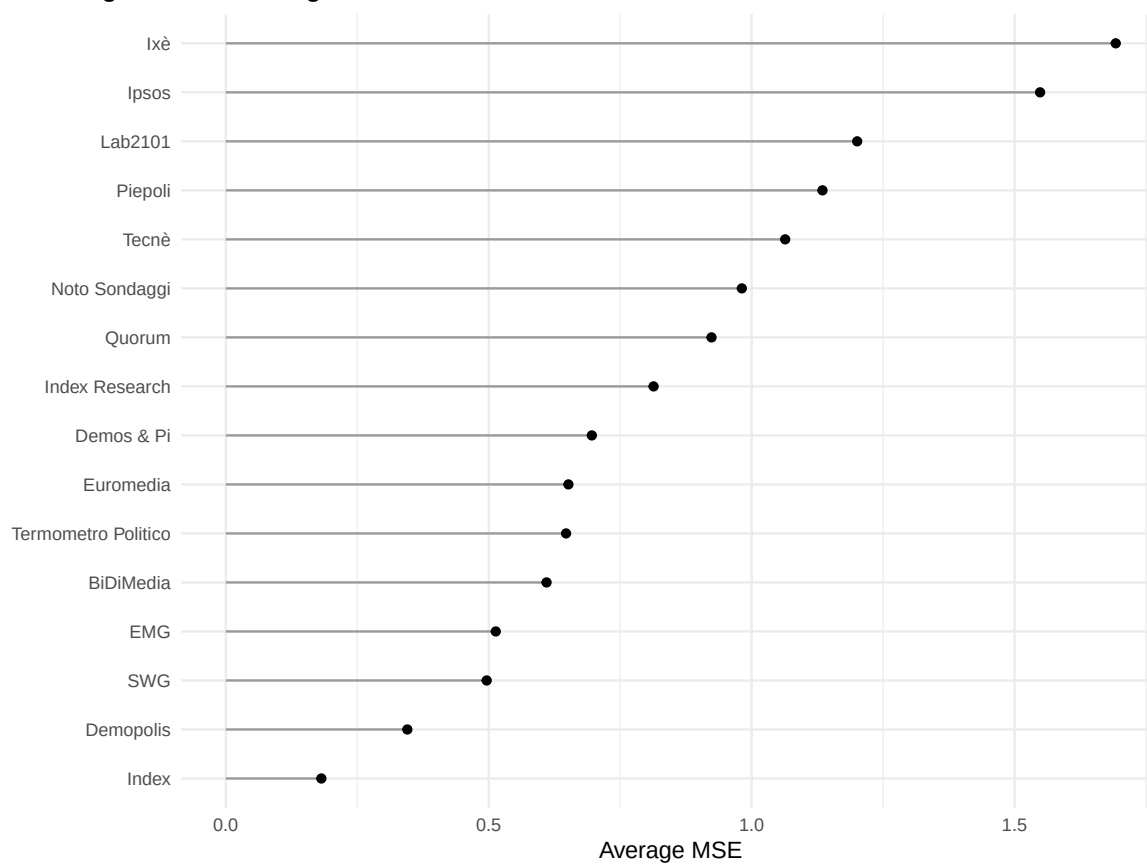


Figure 33: Aggregate ranking by mean squared error across parties (raw, percentage points).

romedia, Demos & Pi, Index and Quorum, lie between the ones with errors that are moderate and consistent with the mixed profiles seen in the party panels. Overall, Graph B provides a compact, single-metric view of how much each institute typically diverges from the cross-party consensus.

Graph C: Mean Absolute Bias (with within-party standardisation)

Construction. For each party I z-standardise bias across institutes, take the absolute value, and then average the resulting normalised values across parties.

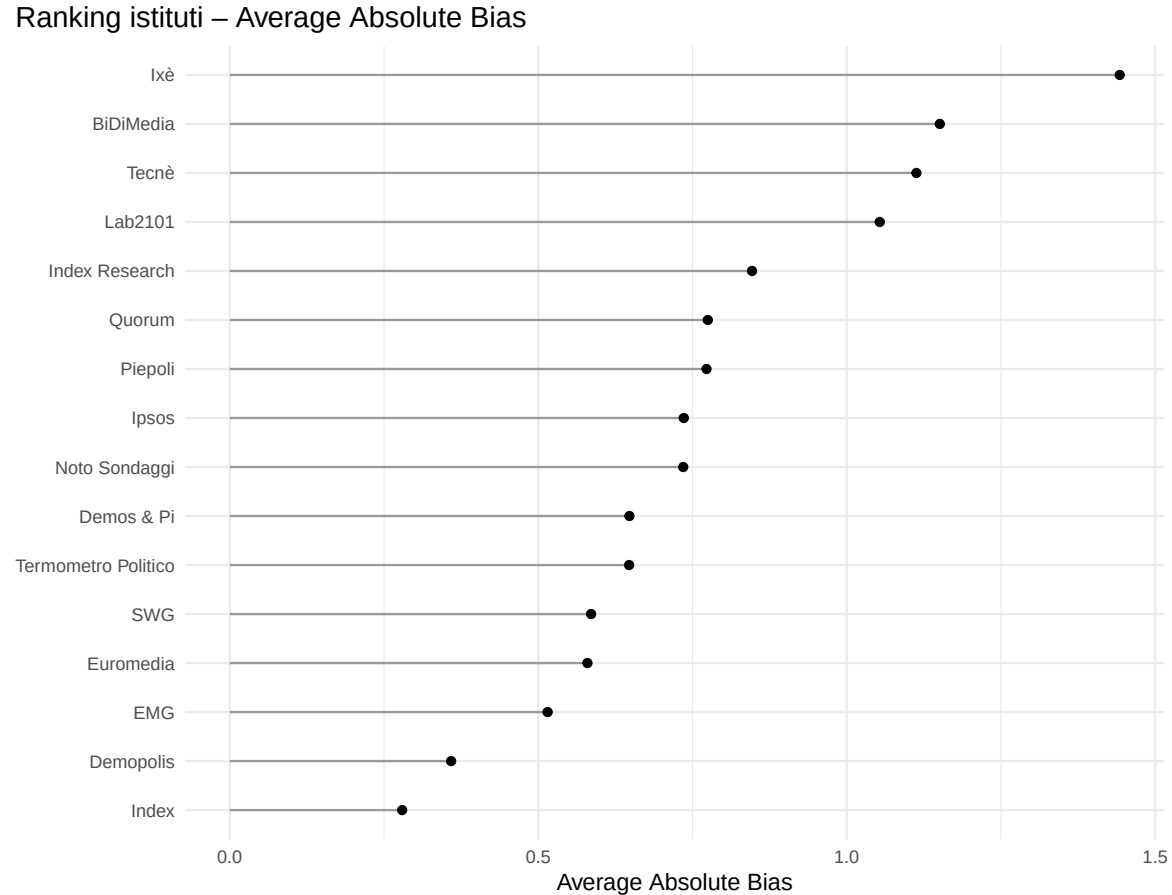


Figure 34: Aggregate ranking by mean absolute bias with within-party z-standardisation (scale-free).

What it shows. Compared with Graph A, the standardised ranking equalises party weights and accentuates relative deviations. Several shifts are visible: BiDiMedia climbs upwards, since its biases on PD respectively appear more extreme once rescaled by within-party variance. Ixè, Lab2101 and Tecnè remain among the worst performers, confirming their structural tilts. Institutes with broadly neutral profiles (Demopolis,

SWG, Euromedia) are pushed further down, as their already small biases become even less distinctive after standardisation. In short, while Graph A reflected absolute magnitudes, Graph C sharpens the contrast by asking which institutes are most deviant relative to their peers for the same party.

Graph D: Mean Squared Error (with within-party standardisation)

Construction. For each party I z-standardise MSE across institutes and then average these z-scores by institute.

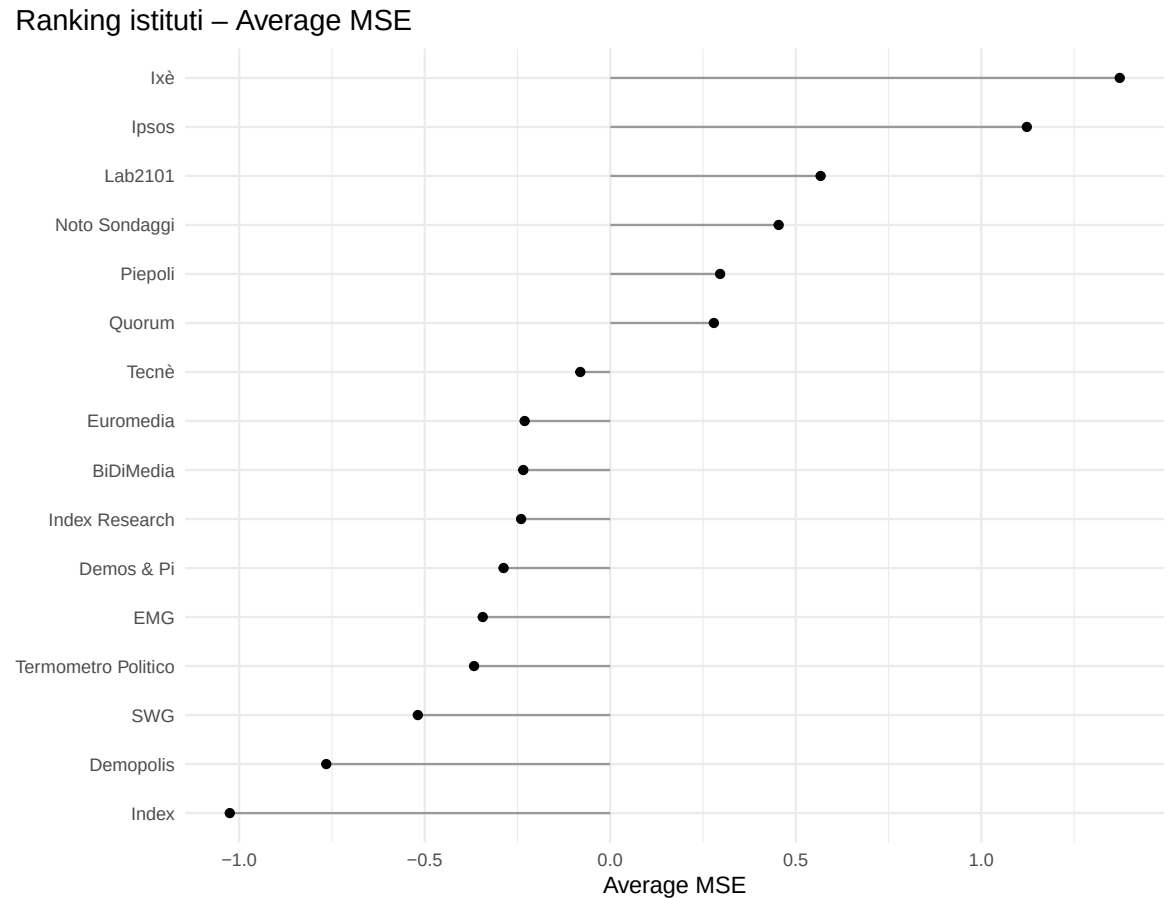


Figure 35: Aggregate ranking by mean squared error with within-party z-standardisation (scale-free).

What it shows. Relative to Graph B, standardisation removes party-level scale effects and highlights persistent outliers. Ixè, Ipsos and Lab2101 remain clearly at the top, confirming that their errors are structurally high across parties. Tecnè moves slightly down than in the raw MSE ranking, because its FI bias is less anomalous once judged relative to the dispersion of other institutes on that party. Demopolis,

SWG and Euromedia stay firmly at the bottom, consistently close to the cross-party trend. Graph D therefore refines Graph B by identifying which institutes are genuine methodological outliers, as opposed to those whose higher raw errors were inflated by scale differences across parties.

Table Analysis: Aggregated institute metrics

Content. Tables 4 and 5 report, for each institute, five summary metrics averaged across the five major parties: mean absolute bias, share of significant biases (95% CI excluding zero), mean signed bias, mean variance, and mean MSE. Table 4 shows the raw values in percentage points (except the share), while Table 5 presents the same statistics after within-party z-standardisation.

Table 4: Aggregated institute metrics (raw, averaged across parties): mean absolute bias, % significant bias, mean signed bias, mean variance, mean MSE. Ordered by MSE (descending).

Institute	Mean Abs. Bias	% Sig. Bias	Mean Bias	Mean Variance	Mean MSE
Ixè	0.716	0.800	0.073	961.590	1.692
Ipsos	0.312	0.600	0.234	1361.656	1.549
Lab2101	0.557	0.400	0.091	700.600	1.201
Piepoli	0.403	0.800	0.015	409.571	1.135
Tecnè	0.586	0.800	0.316	331.217	1.064
Noto Sondaggi	0.303	0.600	-0.132	482.656	0.981
Quorum	0.326	0.600	-0.131	614.609	0.924
Index Research	0.423	0.800	-0.007	446.573	0.814
Demos & Pi	0.334	0.600	0.032	538.490	0.696
Euromedia	0.236	0.800	-0.095	501.039	0.652
Termometro Pol.	0.355	0.800	0.082	1363.625	0.647
BiDiMedia	0.467	1.000	-0.188	636.017	0.610
EMG	0.177	0.400	-0.094	675.927	0.513
SWG	0.286	0.800	-0.096	486.897	0.496
Demopolis	0.165	0.600	0.029	492.564	0.345
Index	0.129	0.600	-0.129	116.155	0.182

Results. The raw figures confirm the main patterns from the cross-party graphs. Ixè stands out with the highest absolute bias (0.72 pp) and one of the largest shares of significant tilts (80%), consistent with its systematic positive bias on PD and negative on Lega. Ipsos records a high MSE (1.55) driven largely by variance (1362), reflecting the noisiness of its estimates despite modest mean bias levels. Lab2101 also

combines large average bias (0.56 pp) with high variance, reproducing the instability already visible in Figure 30. Tecnè registers a moderate mean bias overall (0.59 pp) but with 80% of party-level effects significant (also considering the persistent FI tilt). By contrast, Demopolis and SWG lie at the lower end, with small biases and low MSE values, confirming their broadly neutral behaviour. In between, institutes such as Termometro Politico, BiDiMedia (the only institute that have significant bias for all the parties) and Piepoli display moderate errors with recurrent party-specific tilts, in line with the typology of Figure 31.

Table 5: Aggregated institute metrics (within-party z-standardised, averaged across parties): mean absolute bias, % significant bias, mean signed bias, mean variance, mean MSE. Ordered by MSE (descending).

Institute	Mean Abs. Bias	% Sig. Bias	Mean Bias	Mean Variance	Mean MSE
Ixè	1.443	0.800	0.395	0.708	1.373
Ipsos	0.736	0.600	0.500	1.633	1.123
Lab2101	1.054	0.400	0.062	0.178	0.567
Noto Sondaggi	0.735	0.600	-0.481	-0.351	0.454
Piepoli	0.773	0.800	-0.006	-0.430	0.296
Quorum	0.775	0.600	-0.087	0.066	0.279
Tecnè	1.113	0.800	0.560	-0.490	-0.081
Euromedia	0.580	0.800	-0.372	-0.255	-0.230
BiDiMedia	1.151	1.000	-0.305	0.002	-0.234
Index Research	0.846	0.800	0.062	-0.510	-0.240
Demos & Pi	0.648	0.600	0.169	-0.348	-0.287
EMG	0.515	0.400	-0.392	0.125	-0.344
Termometro Pol.	0.647	0.800	0.169	1.582	-0.367
SWG	0.586	0.800	-0.154	-0.341	-0.518
Demopolis	0.359	0.600	0.159	-0.240	-0.765
Index	0.279	0.600	-0.279	-1.327	-1.025

When moving to the standardised version (Table 5), relative contrasts sharpen. Ixè remains the worst performer, joined by Ipsos at the top: its volatility, already large in raw units, becomes even more anomalous once scaled by within-party dispersion. BiDiMedia also rises, confirming that its raw biases, though moderate in pp terms, are unusually large relative to competitors for PD. Lab2101 continues to score high, consistent with its instability. Conversely, Demopolis and Index consolidate their position at the very bottom, while Euromedia, SWG and EMG cluster as consistently neutral,

variance-driven institutes. Overall, the comparison reinforces the graphical results: a handful of institutes (Ixè, Ipsos, Tecnè, Lab2101) emerge as systematic outliers, while most others remain closer to the average.

3.2.3 Structural dependencies across institutes

Figure 36 summarises, for each pair of institutes, the structural correlations of institute-specific errors estimated by the local-level state-space model with measurement variance

$$H_t = \Sigma_{\text{bias}} + \text{diag}(\sigma_i^2/n_{i,t}).$$

In practice, this shows how much the error components tend to “move together” net of both the common trend and sampling noise. The matrix displayed is the average (via Fisher’s z transform) of the correlations estimated separately for each party.

Each cell measures the association between the errors/biases of two institutes: positive values indicate that when one exhibits a tilt in a given direction for a party, the other tends to show a tilt with the same sign for that party; negative values suggest tilts that are typically opposite.

What we estimate here is not the co-variation driven by sampling noise (accounted for by the term $\text{diag}(\sigma_i^2/n_{i,t})$) or by temporary common fluctuations (captured by the trend). Instead, the matrix Σ_{bias} represents the more stable component of institute differences, for example the part reflecting similar methodological choices (sampling frames, weighting, online vs. telephone mode, commissioning environment) or a shared position within the same “information ecosystem.” Accordingly, a positive correlation between two institutes indicates that, in the long run, their estimation errors tend to shift in the same direction, whereas a negative correlation signals opposite inclinations.

We can highlight how:

1. Dependencies are generally moderate. The picture is dominated by values close to zero, with a few pockets of positive correlation of modest magnitude. This matches the institute-level evidence above: house effects are mostly party-specific

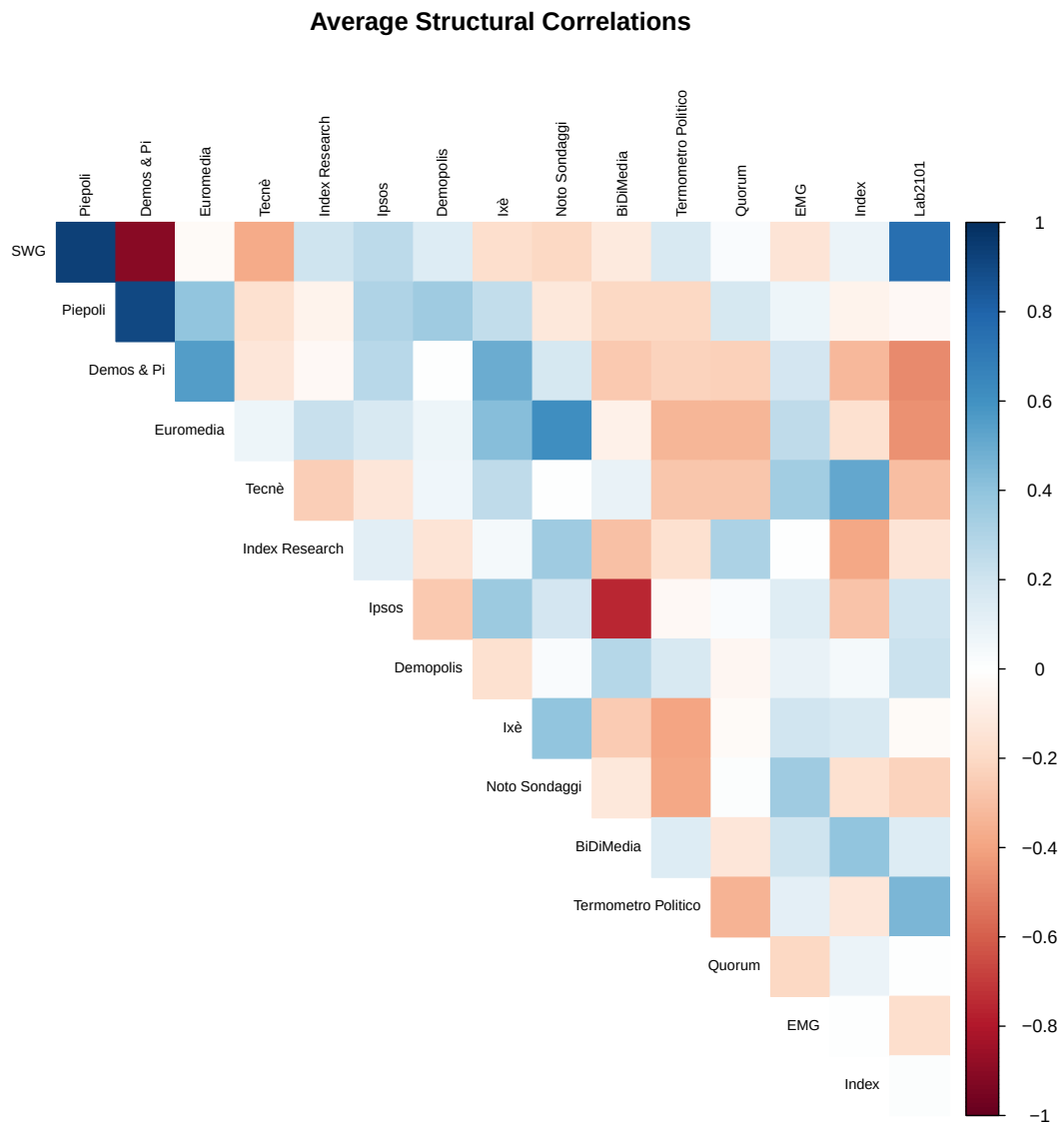


Figure 36: Average structural correlations of institute biases (upper-triangle heatmap). Cells report Fisher- z averaged correlations across the selected parties.

and institute-specific, not driven by a single “pro/against” fracture cutting across the entire industry. This helps explain why, in poll averages, many disagreements are quickly offset: systematic errors are not perfectly aligned across institutes.

2. Micro-clusters rather than opposing “blocks.” The mosaic reveals small groups (pairs/triads) with positive correlations, typical of similar measurement/weighting choices or of similar publication environments. Strongly negative coefficients are comparatively rare. This pattern overlaps with the MSE typology: institutes with low variance and low bias contribute little to dependencies (correlations near zero), whereas those with more persistent tilts tend to form small islands of co-movement.

The evidence on “influences” across institutes confirms the heterogeneity already observed: a few islands of structural co-movement, many weak relationships, and a non-negligible role of dependencies in shaping poll aggregates. This strengthens the case for: (a) explicitly modelling Σ_{bias} ; (b) applying weights and corrections that account for the non-independence of house effects.

4 Conclusions

This thesis has addressed the problem of poll aggregation ("poll of polls") in the Italian context between 2018 and 2024, developing and estimating a family of state-space models that extend the approaches most frequently discussed in the literature. In particular, the models incorporate house effects across institutes, institute-specific heteroskedastic variances linked to both methodological heterogeneity and sample size, non-Gaussian measurement processes (binomial rather than normal), and structural dependencies across polling institutes. The overall objective was not only to obtain a smoother and more robust estimate of the underlying vote intention dynamics, but also to disentangle the sources of divergence among polls, distinguishing the systematic component (bias) from the stochastic component (variance).

The methodological contributions of this work can be summarized as follows. First, the baseline Pelagatti–Monti framework has been extended by including house effects δ_j with a sum-to-zero constraint, institute-specific error variances σ_j^2 scaled by $1/n_{j,t}$, binomial formulations both with and without bias terms, and a structural covariance matrix Σ_{bias} that captures persistent co-movements and methodological affinities among institutes. Second, estimation and inference were performed through Kalman filtering and smoothing for the Gaussian specifications, Durbin-Koopman approximations for the binomial case, and importance sampling for corrected credible intervals, with logarithmic reparametrizations to ensure stability and positivity. Third, interpretable diagnostics were proposed, including a reasonable decomposition of the mean squared error (MSE) into squared bias and average measurement variance, as well as a structural correlation matrix that provides a network-like interpretation of methodological dependencies. Finally, external validation was attempted by comparing the estimated house effects with realized biases at the 2022 general election, showing consistent signs and magnitudes in several cases.

Empirical evidence indicates that variance explains the majority of discrepancies across institutes. For most parties and institutes, MSE is dominated by the

random component driven by sample size and survey design choices. House effects do exist, but they are generally modest in magnitude (around ± 1 percentage point) and strongly party-specific. A few notable patterns emerge: for Lega, some institutes (Lab2101 and Termometro Politico) consistently report higher estimates, while others (Ixè, Demos & Pi, Piepoli) tend to report lower values; for the Partito Democratico, biases are typically smaller; for Fratelli d'Italia, differences are mainly variance-driven; for Movimento 5 Stelle and Forza Italia, dispersion is higher and in the latter case certain institutes (notably Tecnè) show persistent positive deviations. Outlier institutes such as Ixè and Lab2101 combine both variance and systematic tilts, while Ipsos is characterized by large variance, particularly with respect to the Five Star Movement. The estimated correlation structure suggests localized clusters of methodological affinity rather than a single polarized division, which explains why aggregation tends to neutralize many discrepancies. Smoothed party trends confirm well-known dynamics but place them within credible intervals consistent with the available information: the decline of Lega after 2019, the rise of Fratelli d'Italia to electoral leadership in 2022, the relative stability and recovery of the Partito Democratico in the mid-2020s, the contraction and stabilization of the Five Star Movement, and the gradual erosion of Forza Italia.

From a practical standpoint, the results suggest that poll aggregators should weigh surveys not only by recency and sample size, but also by institute-specific variance and systematic house effects. Correcting for persistent biases prevents aggregate estimates from drifting in subtle ways. Equally important is the transparent communication of uncertainty through credible intervals and diagnostics that distinguish between variance-driven and bias-driven discrepancies. The strong heterogeneity in available metadata underscores the importance of systematically publishing sample sizes, design features, and weighting procedures. Finally, the study confirms that the overall quality of the polling ecosystem is critical: aggregation can only partially compensate for weaknesses at the data collection stage.

Several limitations must be acknowledged. The identification of house effects

is relative to the average institute, rather than absolute deviations from the unobserved truth. The local-level state model may underrepresent structural breaks or abrupt campaign shocks, while Σ_{bias} is assumed to be time-invariant. Gaussian approximations may underestimate uncertainty near boundaries, and although importance sampling mitigates this, residual distortions may remain. The analysis was limited to the sixteen most prolific institutes, neglecting smaller players whose practices might differ. Metadata were incomplete for certain institutes, particularly regarding sample sizes, which introduces additional noise. External validation was limited to the 2022 election; ideally, more electoral anchors (other European or municipal elections) would be required. Finally, exogenous factors such as mode of interview, sponsor, and herding behavior were only indirectly accounted for.

Future research may proceed in several directions. A natural extension is to allow house effects δ_j to evolve dynamically, possibly linked to covariates such as interview mode, weighting scheme, sponsor, or survey timing. Multinomial specifications could enforce the simplex constraint across parties, capturing joint vote shifts and substitution patterns. Cross-country replication, beginning with the Austrian dataset already collected, would test the generalizability of findings and highlight differences in media-polling ecosystems. Finally, extensions that translate vote shares into turnout-adjusted seat projections would bring the model closer to institutional outcomes.

In conclusion, this thesis provides (i) a replicable methodological framework, (ii) operational diagnostics for weighting and communicating poll results more responsibly, and (iii) empirical evidence that clarifies when divergences across polls are merely stochastic noise and when they reflect systematic structure. While differences among institutes are real, they are mostly modest and party-specific; random error remains the dominant source of variation. Only a handful of institutes display persistent structural tilts, and identifying them is key to avoiding biased aggregate estimates. Ultimately, responsible aggregation should not be conceived as a simple average, but as a transparent and informed synthesis that integrates bias correction, variance calibration, and structural dependencies into a coherent tool for understanding public opinion.

Appendix Austria

The models developed in this thesis were successfully applied to a second national context, Austria. The methodological transfer is essentially straightforward: it only requires remapping party labels, harmonizing pollster names, and maintaining the same state-space structure (latent trend with heteroskedastic measurement error proportional to $1/n$), both in the Gaussian and in the binomial formulations, with Durbin–Koopman approximations and importance sampling corrections. In this sense, the Austrian analysis serves as a robustness and portability check of the proposed approach.

Voting intentions in Austria

Line = estimated mean; shaded area = 95% credible interval

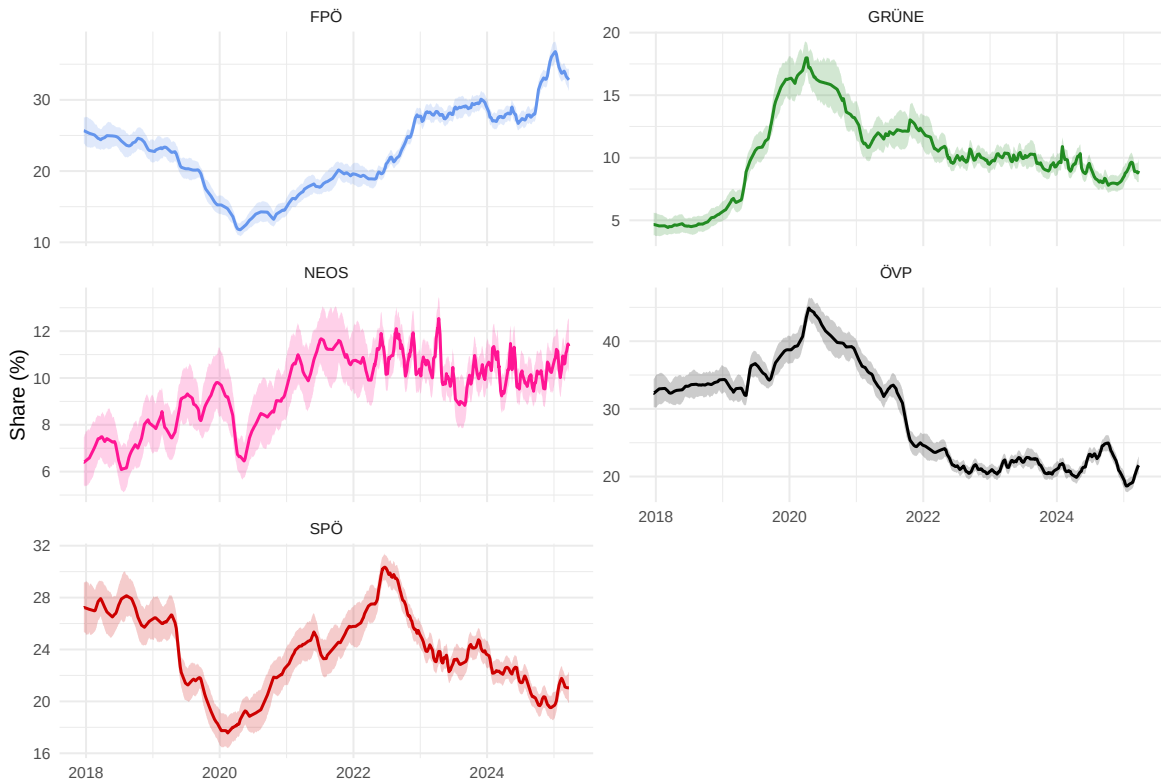


Figure 37: Smoothed voting-intention trends in Austria (2018–2024). The solid line represents the estimated mean vote share for each party, and the shaded areas correspond to 95% credible intervals.

When comparing Austria with the Italian case, both similarities and differences emerge. The main conclusions are consistent across contexts: variance remains the dominant source of discrepancies among polls, while house effects, when present,

Estimated institute bias (house effect)

Points = estimate; bars = 95% credible interval; vertical line = zero bias

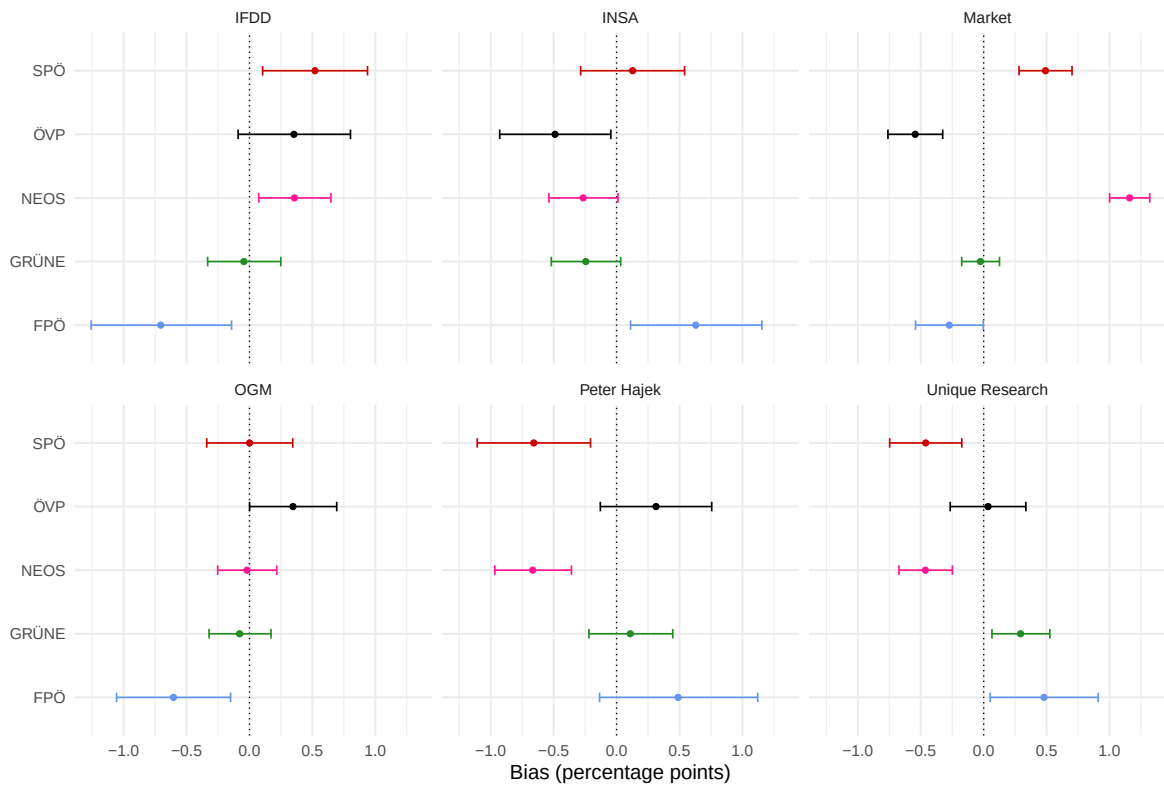


Figure 38: Estimated pollster-specific biases (house effects) by party. Points represent posterior means and horizontal bars the 95% credible intervals; the vertical line indicates zero bias.

are modest and highly party-specific. Overall, the findings confirm that the central recommendations of this thesis variance-based weighting, bias correction, and transparent communication of uncertainty are broadly applicable beyond the Italian case, requiring only minimal operational adjustments.

Mean squared error decomposition by institute

$$\text{MSE} \sim \text{Bias}^2 + \mathbb{E}[s^2 / n]$$

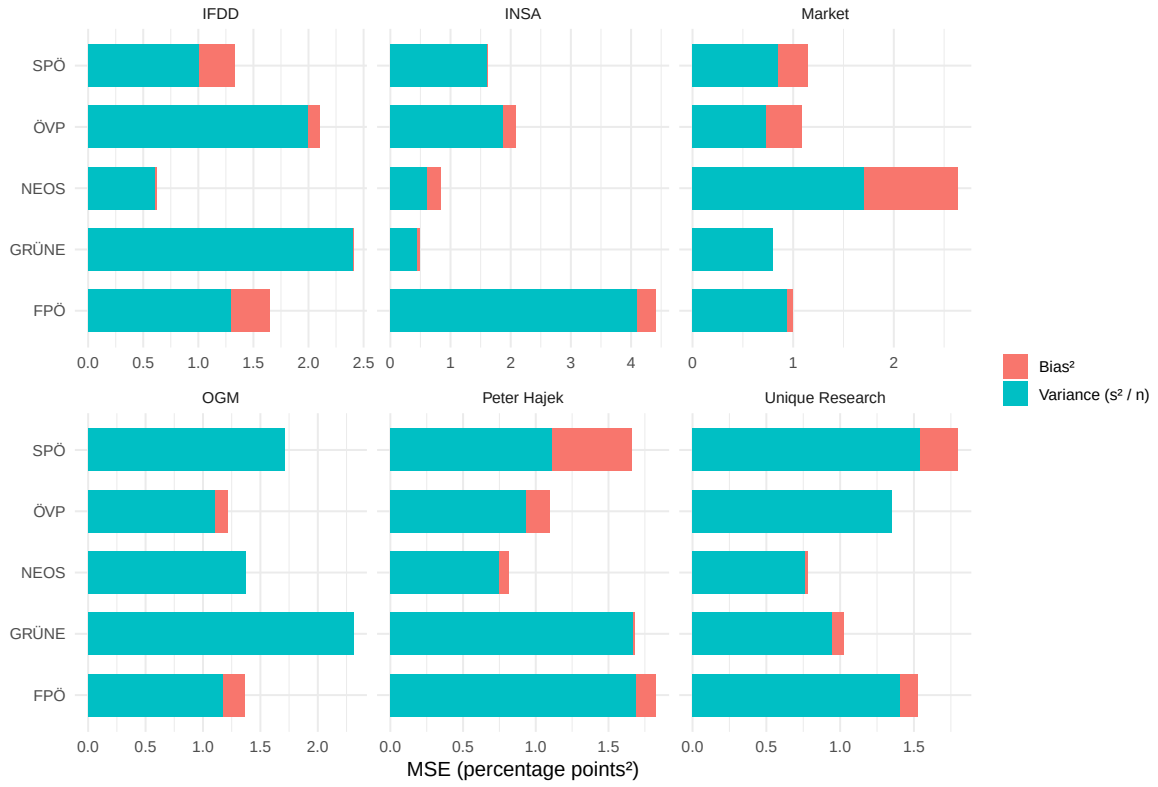


Figure 39: Decomposition of the mean squared error (MSE) by polling institute and party. The MSE is decomposed into its two components: squared bias and expected sampling variance $\mathbb{E}[s^2/n]$.

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